

3 MOSFET Devices

The metal–oxide–semiconductor field-effect transistor (MOSFET) is the building block of VLSI circuits in microprocessors and dynamic memories. Because the current in a MOSFET is transported predominantly by carriers of one polarity only (e.g., electrons in an n-channel device), the MOSFET is usually referred to as a unipolar or majority-carrier device. Throughout this chapter, n-channel MOSFETs are used as an example to illustrate device operation and derive drain-current equations. The results can easily be extended to p-channel MOSFETs by exchanging the dopant types and reversing the voltage polarities.

The basic structure of a MOSFET is shown in Fig. 3.1. It is a four-terminal device with the terminals designated as *gate* (subscript *g*), *source* (subscript *s*), *drain* (subscript *d*), and *substrate* or *body* (subscript *b*). An n-channel MOSFET, or nMOSFET, consists of a p-type silicon substrate into which two n^+ regions, the source and the drain, are formed (e.g., by ion implantation). The gate electrode is usually made of metal or heavily doped polysilicon and is separated from the substrate by a thin silicon dioxide film, the *gate oxide*. The gate oxide is usually formed by thermal oxidation of silicon. In VLSI circuits, a MOSFET is surrounded by a thick oxide called the *field oxide* to isolate it from the adjacent devices. The surface region under the gate oxide between the source and drain is called the *channel* region and is critical for current conduction in a MOSFET. The basic operation of a MOSFET device can be easily understood from the MOS capacitor discussed in Section 2.3. When there is no voltage applied to the gate or when the gate voltage is zero, the p-type silicon surface is either in accumulation or in depletion and there is no current flow between the source and drain. The MOSFET device acts like two back-to-back p–n junction diodes with only low-level leakage currents present. When a sufficiently large positive voltage is applied to the gate, the silicon surface is inverted to n-type, which forms a conducting channel between the n^+ source and drain. If there is a voltage difference between them, an electron current will flow from the source to the drain. A MOSFET device therefore operates like a switch ideally suited for digital circuits. *Since the gate electrode is electrically insulated from the substrate, there is effectively no dc gate current, and the channel is capacitively coupled to the gate via the electric field in the oxide* (hence the name *field-effect transistor*).

3.1 Long-Channel MOSFETs

This section describes the basic characteristics of a long-channel MOSFET, which will serve as the foundation for understanding the more important but more complex

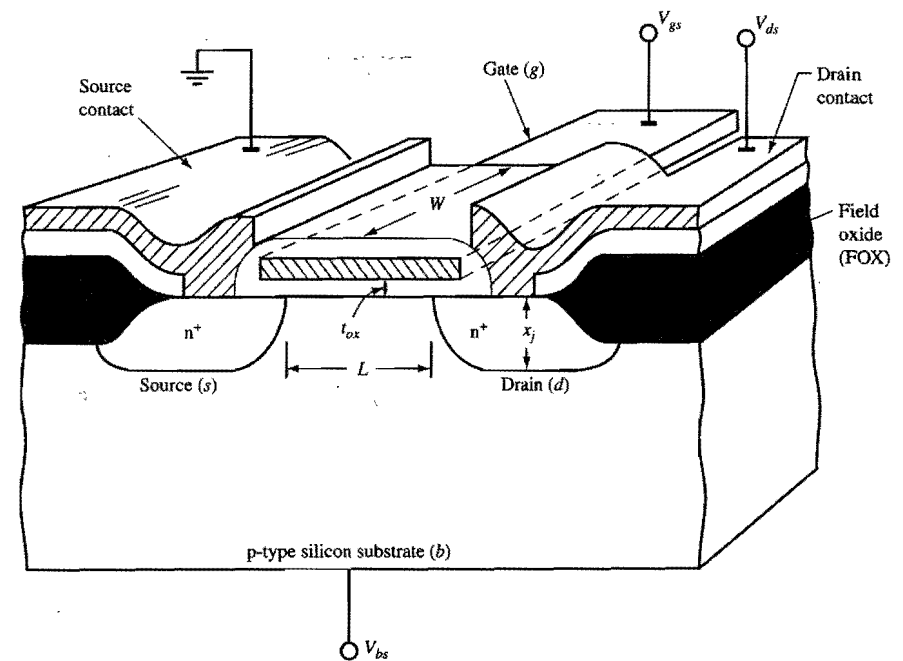


Figure 3.1. Three-dimensional view of basic MOSFET device structure. (After Arora, 1993.)

short-channel MOSFETs in Section 3.2. First, a general MOSFET current model based on the *gradual channel approximation* (GCA) is formulated in Section 3.1.1. The GCA is valid for most regions of MOSFET operation except beyond the pinch-off or saturation point. A *charge-sheet model* is then introduced to obtain implicit equations for the source–drain current. Regional approximations are applied in Section 3.1.2 to derive explicit I – V expressions for the linear and parabolic regions. Current characteristics in the subthreshold region are discussed in Section 3.1.3. Section 3.1.4 addresses the threshold-voltage dependence on substrate bias and temperature. Section 3.1.5 presents an empirical model for electron and hole mobilities in a MOSFET channel. Lastly, intrinsic MOSFET capacitances and inversion-layer capacitance effects (neglected in the regional approximation) are covered in Section 3.1.6.

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3.1.1 Drain-Current Model

In this subsection, we formulate a general drain-current model for a long-channel MOSFET. The model will then be simplified using *charge-sheet approximation*, leading to an analytical expression for the source–drain current. Figure 3.2 shows the schematic cross section of an n-channel MOSFET in which the source is the n^+ region on the left, and the drain is the n^+ region on the right. A thin oxide film separates the gate from the channel region between the source and drain. We choose an x – y coordinate system

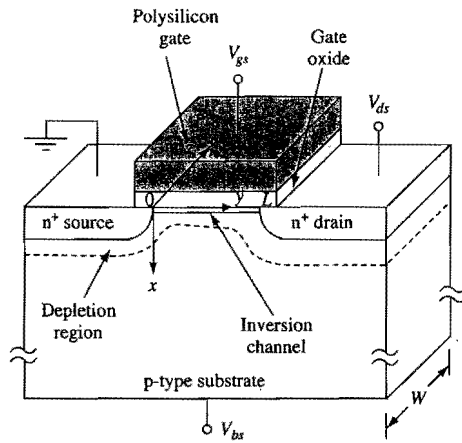


Figure 3.2. A schematic MOSFET cross section, showing the axes of coordinates and the bias voltages at the four terminals for the drain-current model.

consistent with Section 2.3 on MOS capacitors, namely, the x -axis is perpendicular to the gate electrode and is pointing into the p-type substrate with $x=0$ at the silicon surface. The y -axis is parallel to the channel or the current flow direction, with $y=0$ at the source and $y=L$ at the drain. L is called the *channel length* and is a key parameter in a MOSFET device. The MOSFET is assumed to be uniform along the z -axis over a distance called the *channel width*, W , determined by the boundaries of the thick field oxide.

Conventionally, the source voltage is defined as the ground potential. The drain voltage is V_{ds} , the gate voltage is V_{gs} , and the p-type substrate is biased at V_{bs} . Initially, we assume $V_{bs}=0$, i.e., the substrate contact is grounded to the source potential. Later on, we will discuss the effect of substrate bias on MOSFET characteristics. The p-type substrate is assumed to be uniformly doped with an acceptor concentration N_a .

3.1.1.1 Gradual-Channel Approximation

One of the key assumptions in any 1-D MOSFET model is the **gradual channel approximation (GCA), which assumes that the variation of the electric field in the y -direction (along the channel) is much less than the corresponding variation in the x -direction (perpendicular to the channel)** (Pao and Sah, 1966). This allows us to reduce the 2-D Poisson equation to 1-D slices (x -component only) as in Eq. (2.175). The GCA is valid for most of the channel regions except beyond the *pinch-off* point, which will be discussed later.

As defined in Section 2.3.2, $\psi(x, y)$ is the band bending, or intrinsic potential, at (x, y) with respect to the intrinsic potential of the bulk substrate. We further assume that $V(y)$ is the electron quasi-Fermi potential at a point y along the channel with respect to the Fermi potential of the n^+ source. The assumption that V is independent of x in the direction perpendicular to the surface is justified by the consideration that current is proportional to

the gradient of the quasi-Fermi potential and that the MOSFET current flows predominantly in the source-to-drain, or y -direction. At the source end of the channel, $V(y=0)=0$. At the drain end of the channel, $V(y=L)=V_{ds}$, the reverse bias of the drain-to-substrate junction since $V_{bs}=0$. For a vertical slice between the source and drain, the channel-to-substrate diode is reverse biased at $V(y)$ which plays the same role as V_R in Section 2.3.5 on MOS capacitors under nonequilibrium. As depicted in Fig. A4.5 for a reverse biased p-n junction, the electron quasi-Fermi potential is essentially flat in the vertical direction across the n-type inversion layer, and is displaced by $V(y)$ from the Fermi potential of the p-type substrate.

From Eq. (2.178) and Eq. (2.214), the electron concentration at any point (x, y) is given by

$$n(x, y) = \frac{n_i^2}{N_a} e^{q(\psi - V)/kT}. \quad (3.1)$$

Following the same approach as in Section 2.3.2, one obtains an expression for the electric field similar to that of Eq. (2.181):

$$\mathcal{E}^2(x, y) = \left(\frac{d\psi}{dx}\right)^2 = \frac{2kT N_a}{\epsilon_{si}} \left[\left(e^{-q\psi/kT} + \frac{q\psi}{kT} - 1 \right) + \frac{n_i^2}{N_a^2} \left(e^{-qV/kT} (e^{q\psi/kT} - 1) - \frac{q\psi}{kT} \right) \right]. \quad (3.2)$$

The condition for surface inversion, Eq. (2.217), becomes

$$\psi(0, y) = V(y) + 2\psi_B, \quad (3.3)$$

which is a function of y . From Eq. (2.218), the maximum depletion layer width is

$$W_{dm}(y) = \sqrt{\frac{2\epsilon_{si}[V(y) + 2\psi_B]}{qN_a}}, \quad (3.4)$$

which is also a function of y .

3.1.1.2 Pao and Sah's Double Integral

Under the assumption that both the hole current and the generation and recombination current are negligible, the current continuity equation can be applied to the electron current in the y -direction. In other words, the total drain-to-source current I_{ds} is the same at any point along the channel. From Eq. (2.63), the electron current density at a point (x, y) is

$$J_n(x, y) = -q\mu_n n(x, y) \frac{dV(y)}{dy}, \quad (3.5)$$

where $n(x, y)$ is the electron density, and μ_n is the electron mobility in the channel. The carrier mobility in the channel is generally much lower than the mobility in the bulk, due to additional surface scattering mechanisms, as will be addressed in

Section 3.1.5. With $V(y)$ defined as the quasi-Fermi potential, i.e., playing the role of ϕ_n in Eq. (2.63), Eq. (3.5) includes both the drift and diffusion currents. The total current at a point y along the channel is obtained by multiplying Eq. (3.5) with the channel width W and integrating over the depth of the current-carrying layer. The integration is carried out from $x=0$ to x_i , where x_i is a depth into the p-type substrate but not infinity.¹

$$I_{ds}(y) = qW \int_0^{x_i} \mu_n n(x, y) \frac{dV}{dy} dx. \quad (3.6)$$

There is a sign change, as we define $I_{ds} > 0$ to be the drain-to-source current in the $-y$ direction. Since V is a function of y only, dV/dy can be taken outside the integral. We also assume that μ_n can be taken outside the integral by defining an effective mobility, μ_{eff} , at some average gate and drain fields. What remains in the integral is the electron concentration, $n(x, y)$. Its integration over the inversion layer gives the inversion charge per unit gate area, Q_i :

$$Q_i(y) = -q \int_0^{x_i} n(x, y) dx. \quad (3.7)$$

Equation (3.6) then becomes

$$I_{ds}(y) = -\mu_{eff} W \frac{dV}{dy} Q_i(y) = -\mu_{eff} W \frac{dV}{dy} Q_i(V). \quad (3.8)$$

In the last step, Q_i is expressed as a function of V ; V is interchangeable with y , since V is a function of y only. Multiplying both sides of Eq. (3.8) by dy and integrating from 0 to L (source to drain) yield

$$\int_0^L I_{ds} dy = \mu_{eff} W \int_0^{V_{ds}} [-Q_i(V)] dV. \quad (3.9)$$

Current continuity requires that I_{ds} be a constant, independent of y . Therefore, the drain-to-source current is

$$I_{ds} = \mu_{eff} \frac{W}{L} \int_0^{V_{ds}} [-Q_i(V)] dV. \quad (3.10)$$

An alternative form of $Q_i(V)$ can be derived if $n(x, y)$ is expressed as a function of (ψ, V) using Eq. (3.1), i.e.,

$$n(x, y) = n(\psi, V) = \frac{n_i^2}{N_a} e^{q(\psi-V)/kT}, \quad (3.11)$$

¹ It does not matter what the exact choice of x_i is. See further discussions after Eq. (3.12).

and substituted into Eq. (3.7):

$$\begin{aligned} Q_i(V) &= -q \int_{\psi_s}^{\delta} n(\psi, V) \frac{dx}{d\psi} d\psi \\ &= -q \int_{\delta}^{\psi_s} \frac{(n_i^2/N_a) e^{q(\psi-V)/kT}}{\mathcal{E}(\psi, V)} d\psi. \end{aligned} \quad (3.12)$$

Here, ψ_s is the surface potential at $x=0$ and $\mathcal{E}(\psi, V) = -d\psi/dx$ is given by the square root of Eq. (3.2). The lower integration limit, δ , represents any small potential $\ll kT/q$, but not zero as the integral is unbounded at $\psi = 0$.² Substituting Eq. (3.12) into Eq. (3.10) yields

$$I_{ds} = q\mu_{eff} \frac{W}{L} \int_0^{V_{ds}} \left(\int_{\delta}^{\psi_s} \frac{(n_i^2/N_a) e^{q(\psi-V)/kT}}{\mathcal{E}(\psi, V)} d\psi \right) dV. \quad (3.13)$$

This is referred to as *Pao and Sah's double integral* (Pao and Sah, 1966). The boundary value ψ_s is determined by two coupled equations: Eq. (2.195) and $Q_s = -\epsilon_{si} \mathcal{E}_s(\psi_s)$ or Gauss's law, where $\mathcal{E}_s(\psi_s)$ is obtained by letting $\psi = \psi_s$ in Eq. (3.2). In depletion and inversion where $q\psi_s/kT \gg 1$, only two of the terms in Eq. (3.2) are significant and need to be kept. The merged equation is then

$$\begin{aligned} V_{gs} &= V_{fb} + \psi_s - \frac{Q_s}{C_{ox}} \\ &= V_{fb} + \psi_s + \frac{\sqrt{2\epsilon_{si}kTN_a}}{C_{ox}} \left[\frac{q\psi_s}{kT} + \frac{n_i^2}{N_a^2} e^{q(\psi_s-V)/kT} \right]^{1/2}, \end{aligned} \quad (3.14)$$

which is an implicit equation for $\psi_s(V)$. Equations (3.14) and (3.13) can only be solved numerically.

3.1.1.3 Charge-Sheet Model

Pao and Sah's double integral can be simplified to a single integral if the inversion charge density Q_i can be expressed as a function of ψ_s . This is accomplished by the *charge-sheet model* (Brews, 1978) which is based on the fact that the inversion layer is located very close to the silicon surface like a thin sheet of charge. There is an abrupt increase of the field (spatial integration of volume charge density) across the thin inversion layer, but very little change of the potential (spatial integration of field). As shown in the example in Fig. 2.36, neither the surface potential nor the depletion charge density changes much after strong inversion. The central assumption of the charge-sheet model is that Eq. (2.189) for the depletion charge density,

$$Q_d = -qN_a W_d = -\sqrt{2\epsilon_{si}qN_a\psi_s}, \quad (3.15)$$

can be extended to beyond strong inversion. (Actually, once the inversion charge dominates, Q_d hardly changes with ψ_s . See Ex. 2.6.) Since the total silicon charge density

² For very small ψ , the integral yields Q_i/q of $(n_i^2/N_a)L_D \ln \psi$ where L_D is the Debye length (Eq. (2.53)). This is many orders of magnitude below the level of interest even for a factor of 10^{10} change in ψ . But $\psi = 0$ is equivalent to $x_i = \infty$ in Eq. (3.7), in which case Q_i diverges.

Q_s is given by Eq. (3.14) or Eq. (2.195), Eq. (3.15) allows the inversion charge density to be expressed as

$$Q_i = Q_s - Q_d = -C_{ox}(V_{gs} - V_{fb} - \psi_s) + \sqrt{2\epsilon_{si}qN_a\psi_s}. \quad (3.16)$$

It should be noted that the charge sheet model does not literally assume all the inversion charge is located at the silicon surface with a zero depth. That would mean $dQ_i/dV_{gs} = C_{ox}$, which is not the case with Eq. (3.16) since ψ_s also increases with V_{gs} as described by Eq. (3.14).

The variable in the drain current integral, Eq. (3.10), can be transformed from V to ψ_s ,

$$I_{ds} = \mu_{eff} \frac{W}{L} \int_{\psi_{s,s}}^{\psi_{s,d}} (-Q_i(\psi_s)) \frac{dV}{d\psi_s} d\psi_s, \quad (3.17)$$

where $\psi_{s,s}$, $\psi_{s,d}$ are the values of the surface potential at the source and the drain ends of the channel. For given V_{gs} and V_{ds} , they can be solved numerically from the implicit equation (3.14) by setting $V = 0$ (for $\psi_{s,s}$) and $V = V_{ds}$ (for $\psi_{s,d}$), respectively. Equation (3.14) can also be used to solve for $V(\psi_s)$,

$$V = \psi_s - \frac{kT}{q} \ln \left\{ \frac{N_a^2}{n_i^2} \left[\frac{C_{ox}^2 (V_{gs} - V_{fb} - \psi_s)^2}{2\epsilon_{si}kTN_a} - \frac{q\psi_s}{kT} \right] \right\}, \quad (3.18)$$

and evaluate its derivative:

$$\frac{dV}{d\psi_s} = 1 + \frac{2kT}{q} \frac{C_{ox}^2 (V_{gs} - V_{fb} - \psi_s) + \epsilon_{si}qN_a}{C_{ox}^2 (V_{gs} - V_{fb} - \psi_s)^2 - 2\epsilon_{si}qN_a\psi_s}. \quad (3.19)$$

Substituting Eqs. (3.16) and (3.19) into Eq. (3.17) yields

$$I_{ds} = \mu_{eff} \frac{W}{L} \int_{\psi_{s,s}}^{\psi_{s,d}} \left[C_{ox}(V_{gs} - V_{fb} - \psi_s) - \sqrt{2\epsilon_{si}qN_a\psi_s} + \frac{2kT}{q} \frac{C_{ox}^2 (V_{gs} - V_{fb} - \psi_s) + \epsilon_{si}qN_a}{C_{ox}^2 (V_{gs} - V_{fb} - \psi_s)^2 - 2\epsilon_{si}qN_a\psi_s} \right] d\psi_s, \quad (3.20)$$

which expresses the drain current in a single integral.

It is too tedious to carry out the integral in Eq. (3.20) exactly. A second approximation is introduced in the charge sheet model (Brews, 1978) to obtain an analytical expression for the drain current. Note that the first two terms in the square bracket of Eq. (3.20) is simply $-Q_i$. Because of the kT/q multiplier, the last term in the square brackets is usually much smaller than the first two unless $Q_i \approx 0$ which happens when $C_{ox}(V_{gs} - V_{fb} - \psi_s) \approx \sqrt{2\epsilon_{si}qN_a\psi_s}$. It is then a good approximation to apply this relation to the last term in the square brackets so that the integral can be carried out analytically:

$$I_{ds} = \mu_{eff} \frac{W}{L} \left\{ C_{ox} \left(V_{gs} - V_{fb} + \frac{kT}{q} \right) \psi_s - \frac{1}{2} C_{ox} \psi_s^2 - \frac{2}{3} \sqrt{2\epsilon_{si}qN_a\psi_s}^3 + \frac{kT}{q} \sqrt{2\epsilon_{si}qN_a\psi_s} \right\} \bigg|_{\psi_{s,s}}^{\psi_{s,d}}. \quad (3.21)$$

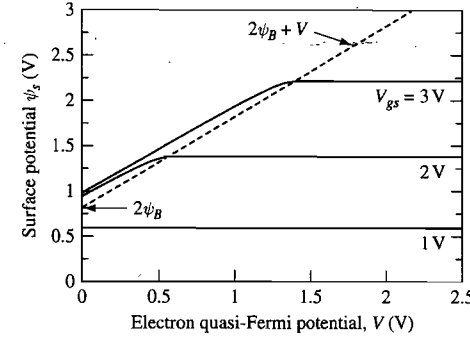


Figure 3.3. Numerical solutions of the implicit Eq. (3.14) for three values of V_{gs} . The dotted line represents the regional approximation used in Section 3.1.2. The MOS device parameters are $N_a = 10^{17} \text{ cm}^{-3}$, $t_{ox} = 10 \text{ nm}$, and $V_{fb} = 0$.

Because Eq. (3.21) covers all regions of MOSFET operation: subthreshold, linear, and saturation in a single, continuous function, it has become the basis of all surface potential based compact models for circuit (SPICE) simulations (Gildenblat *et al.*, 2006). Many numerical methods have been developed to solve the implicit Eq. (3.14) for $\psi_{s,s}$ and $\psi_{s,d}$ given V_{gs} and V_{ds} . They employ either explicit approximations or iterative procedures. The general behavior of the solution $\psi_s(V)$ is shown in an example in Fig. 3.3. The device parameters are the same as those of Fig. 2.36. For $V_{gs} = 1 \text{ V}$, the device is below threshold where the inversion charge is negligible, i.e., the $(n_i^2/N_a^2)e^{q(\psi_s-V)/kT}$ term in the square brackets in Eq. (3.14) is negligible. The solution ψ_s depends on V_{gs} (see Fig. 2.36), but is totally insensitive to V . For $V_{gs} = 2 \text{ V}$, the MOSFET is turned on. Here, ψ_s increases more or less linearly with V when V is not too large. As V increases (for large enough V_{ds}), ψ_s reaches a saturation value beyond which it becomes independent of V . This is called the *pinch-off* condition where Q_i given by Eq. (3.16) becomes very small. The argument of the log function in Eq. (3.18) also approaches zero. For $V_{gs} = 3 \text{ V}$, the saturation value of ψ_s increases while the saturation happens at a higher V (or V_{ds}).

Because of the two simplifying approximations used, the current calculated from the charge sheet model, Eq. (3.21), deviates from that of Pao and Sah's double integral, Eq. (3.13). The error is a function of doping concentration, oxide thickness, gate and drain bias voltages. Typically, it can be of the order of 10% under certain conditions when biased above threshold (Kyung, 2005). The error is generally larger in subthreshold where the current levels are low and high accuracy is not a paramount issue.

3.1.2

MOSFET I - V Characteristics

In this subsection, we derive the basic I - V characteristics of a long-channel MOSFET in the linear and parabolic regions.

3.1.2.1 Regional Approximations

To obtain explicit equations for the drain current, it is necessary to apply regional approximations to break the charge-sheet model into piecewise models. After the onset of inversion but before saturation, the surface potential can be approximated by $\psi_s = 2\psi_B + V(y)$, or Eq. (3.3). This relation is plotted in Fig. 3.3 (dotted line) for comparison with the more exact curves. It then follows that $dV/d\psi_s = 1$ and Eq. (3.17) can be readily integrated. Applying $\psi_{s,s} = 2\psi_B$ and $\psi_{s,d} = 2\psi_B + V_{ds}$, we obtain the drain current as a function of the gate and drain voltages:

$$I_{ds} = \mu_{eff} C_{ox} \frac{W}{L} \left\{ \left(V_{gs} - V_{fb} - 2\psi_B - \frac{V_{ds}}{2} \right) V_{ds} - \frac{2\sqrt{2\epsilon_{si}qN_a}\psi_B}{3C_{ox}} \left[(2\psi_B + V_{ds})^{3/2} - (2\psi_B)^{3/2} \right] \right\}. \quad (3.22)$$

Equation (3.22) represents the basic I - V characteristics of a MOSFET device based on the charge-sheet model. It indicates that, for a given V_{gs} , the drain current I_{ds} first increases linearly with the drain voltage V_{ds} (called the *linear* or *triode* region), then gradually levels off to a saturated value (*parabolic* region). These two distinct regions are further examined below.

3.1.2.2 Characteristics in the Linear (Triode) Region

When V_{ds} is small, one can expand Eq. (3.22) into a power series in V_{ds} and keep only the lowest-order (first-order) terms:

$$\begin{aligned} I_{ds} &= \mu_{eff} C_{ox} \frac{W}{L} \left(V_{gs} - V_{fb} - 2\psi_B - \frac{\sqrt{4\epsilon_{si}qN_a}\psi_B}{C_{ox}} \right) V_{ds} \\ &= \mu_{eff} C_{ox} \frac{W}{L} (V_{gs} - V_t) V_{ds}, \end{aligned} \quad (3.23)$$

where V_t is the *threshold voltage* given by

$$V_t = V_{fb} + 2\psi_B + \frac{\sqrt{4\epsilon_{si}qN_a}\psi_B}{C_{ox}}. \quad (3.24)$$

Comparing this equation with Eq. (2.202), one can see that V_t is *simply the gate voltage when the surface potential or band bending reaches $2\psi_B$ and the silicon charge (the square root) is equal to the bulk depletion charge for that potential*. As a reminder, $2\psi_B = (2kT/q) \ln(N_a/n_i)$, which is typically 0.6–0.9 V. When V_{gs} is below V_t , there is very little current flow and the MOSFET is said to be in the *subthreshold* region, to be discussed in Section 3.1.3. Equation (3.23) indicates that, *in the linear region, the MOSFET simply acts like a resistor with a sheet resistivity, $\rho_{sh} = 1/[\mu_{eff} C_{ox} (V_{gs} - V_t)]$, modulated by the gate voltage*. The threshold voltage V_t can be determined by plotting I_{ds} versus V_{gs} at low drain voltages, as shown in Fig. 3.4. The extrapolated intercept of the linear portion of the $I_{ds}(V_{gs})$ curve with the V_{gs} -axis gives the approximate value of V_t . In reality, such a *linearly extrapolated threshold voltage* (V_{on}) is slightly

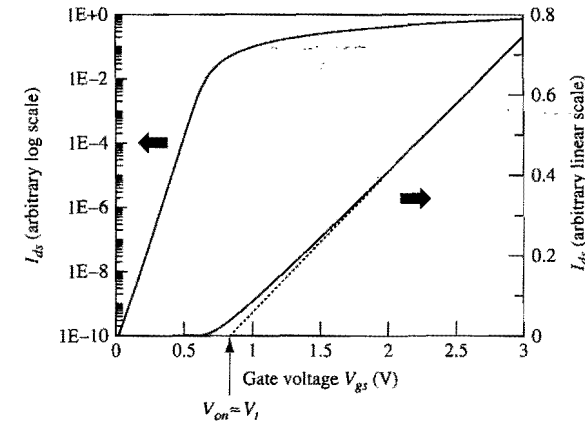


Figure 3.4. Typical MOSFET I_{ds} - V_{gs} characteristics at low drain bias voltages. The same current is plotted on both linear and logarithmic scales. The dotted line illustrates the determination of the linearly extrapolated threshold voltage, V_{on} .

higher than the " $2\psi_B$ " V_t due to inversion-layer capacitance and other effects, as seen in Fig. 2.36 and further addressed in Section 3.1.6. Low-drain $I_{ds}(V_{gs})$ curves are also used to extract the *effective channel length* of a MOSFET, which is discussed in Chapter 4.

3.1.2.3 Characteristics in the Parabolic Region

For larger values of V_{ds} , the second-order terms in the power series expansion of Eq. (3.22) are also important and must be kept. A good approximation to the drain current is then

$$I_{ds} = \mu_{eff} C_{ox} \frac{W}{L} \left((V_{gs} - V_t) V_{ds} - \frac{m}{2} V_{ds}^2 \right), \quad (3.25)$$

where

$$m = 1 + \frac{\sqrt{\epsilon_{si}qN_a/4\psi_B}}{C_{ox}} \quad (3.26)$$

is a factor greater than one, which is related to the *subthreshold slope* and the *body effect* to be discussed in Subsections 3.1.3 and 3.1.4. Equation (3.26) can be converted to several alternative expressions by using Eq. (2.201) for the bulk depletion capacitance C_{dm} at $\psi_s = 2\psi_B$:

$$m = 1 + \frac{C_{dm}}{C_{ox}} = 1 + \frac{3t_{ox}}{W_{dm}}. \quad (3.27)$$

The last expression follows from $C_{dm} = \epsilon_{si}/W_{dm}$, $C_{ox} = \epsilon_{ox}/t_{ox}$, and $\epsilon_{si}/\epsilon_{ox} \approx 3$. A graphical interpretation of m is given in Fig. 3.5. At the threshold condition, $\psi_s = 2\psi_B$, the MOSFET acts like two capacitors, C_{ox} and C_{dm} , in series as the inversion charge capacitance is still

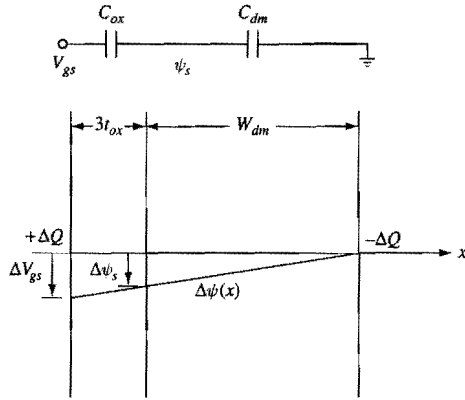


Figure 3.5. Incremental change of potential in a MOSFET due to a gate-voltage modulation near or below threshold. Grounding of the body anchored the potential on the bulk side of the depletion region where $\Delta\psi = 0$. The potential drop across the oxide, $(\Delta Q/\epsilon_{ox})t_{ox}$, is equivalent to $(\Delta Q/\epsilon_{si})[(\epsilon_{si}/\epsilon_{ox})t_{ox}]$. The factor m is defined as $\Delta V_{gs}/\Delta\psi_s$, which equals $(W_{dm} + 3t_{ox})/W_{dm}$.

negligible. The factor m equals $\Delta V_{gs}/\Delta\psi_s$, where $\Delta\psi_s$ is the incremental change of surface potential due to ΔV_{gs} , an incremental change of gate voltage. ΔV_{gs} induces sheet charge densities $+\Delta Q$ at the gate and $-\Delta Q$ at the far edge of the depletion region. They cause a field change of $\Delta\mathcal{E} = \Delta Q/\epsilon_{si}$ in the silicon and $\Delta Q/\epsilon_{ox}$ in the oxide, which give rise to an incremental change of potential $\Delta\psi(x)$ as shown in Fig. 3.5. Here, the oxide width is expanded to $\epsilon_{si}/\epsilon_{ox} \approx 3$ times its physical width so there is no change of slope at the silicon-oxide interface. **While Eq. (3.26) is only valid for uniform bulk doping, Eq. (3.27) is more generally valid for nonuniform doping profiles** to be discussed in Section 4.2.2. Since $1/m$ is a measure of the efficiency of the gate in modulating the surface potential, m should be kept close to one, e.g., between 1.1 and 1.4, in MOSFET design.

Equation (3.25) indicates that as V_{ds} increases, I_{ds} follows a parabolic curve, as shown in Fig. 3.6, until a maximum or saturation value is reached. This occurs when $V_{ds} = V_{dsat} = (V_{gs} - V_t)/m$, at which

$$I_{ds} = I_{dsat} = \mu_{eff} C_{ox} \frac{W(V_{gs} - V_t)^2}{2mL} \quad (3.28)$$

Equation (3.28) reduces to the well-known expression for the MOSFET saturation current when the bulk depletion charge is neglected (valid for low substrate doping) so $m = 1$. The dashed curve in Fig. 3.6 shows the trajectory of V_{dsat} through the various $I_{ds} - V_{ds}$ curves for different V_{gs} . Because of the regional approximation, $\psi_s = 2\psi_B + V$, used in the derivation, Eq. (3.22) and therefore Eq. (3.25) are valid only for $V_{ds} \leq V_{dsat}$. Beyond V_{dsat} , one must go back to the more general Eq. (3.21) coupled with Eq. (3.14). Since $\psi_{s,d}$ saturates at large V_{ds} as depicted in Fig. 3.3, I_{ds} stays constant at I_{dsat} , independent of V_{ds} for $V_{ds} \geq V_{dsat}$.

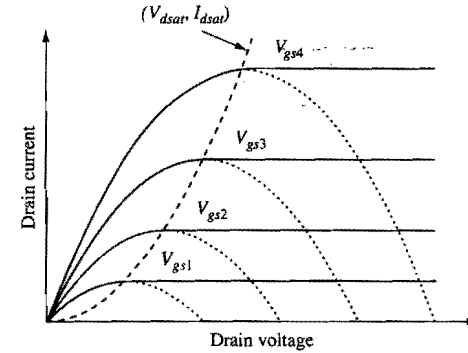


Figure 3.6. Long-channel MOSFET $I_{ds} - V_{ds}$ characteristics (solid curves) for several different values of V_{gs} . The dashed curve shows the trajectory of drain voltage beyond which the current saturates. The dotted curves help to illustrate the parabolic behavior of the characteristics before saturation.

3.1.2.4 The Onset of Pinch-Off and Current Saturation

The saturation of drain current can be understood from the inversion charge density, Eq. (3.16). For $\psi_s = 2\psi_B + V$ and $V \leq 2\psi_B$, one can expand the square-root term of Eq. (3.16) into a power series in V and keep only the two lowest terms,

$$Q_i(V) = -C_{ox}(V_{gs} - V_t - mV). \quad (3.29)$$

$Q_i(V)$ is plotted in Fig. 3.7. Equation (3.10) states that the drain current is proportional to the area under the $-Q_i(V)$ curve between $V = 0$ and V_{ds} . When V_{ds} is small (linear region), the inversion charge density at the drain end of the channel is only slightly lower than that at the source end. As the drain voltage increases (for a fixed gate voltage), the current increases, but the inversion charge density at the drain decreases until finally it goes to zero when $V_{ds} = V_{dsat} = (V_{gs} - V_t)/m$. At this point, I_{ds} reaches its maximum value. In other words, **the surface channel vanishes at the drain end of the channel when saturation occurs**. This is called *pinch-off* and is illustrated in Fig. 3.8. When V_{ds} increases beyond saturation, the pinch-off point moves toward the source, but the drain current remains essentially the same. This is because for $V_{ds} > V_{dsat}$, the voltage at the pinch-off point remains at V_{dsat} and the current, given by

$$\int_0^{L'} I_{ds} dy = \mu_{eff} W \int_0^{V_{dsat}} [-Q_i(V)] dV, \quad (3.30)$$

stays the same apart from a slight decrease in L (to L'), as shown in Fig. 3.8. This phenomenon is called *channel length modulation* and will be discussed in association with short-channel MOSFETs in Section 3.2.

Further insight into the MOSFET behavior at pinch-off can be gained by examining the function $V(y)$. Integrating from 0 to y after multiplying both sides of Eq. (3.8) by dy yields

$$\begin{aligned} I_{ds} y &= \mu_{eff} W \int_0^V [-Q_i(V)] dV \\ &= \mu_{eff} C_{ox} W \left[(V_{gs} - V_t)V - \frac{m}{2} V^2 \right], \end{aligned} \quad (3.31)$$

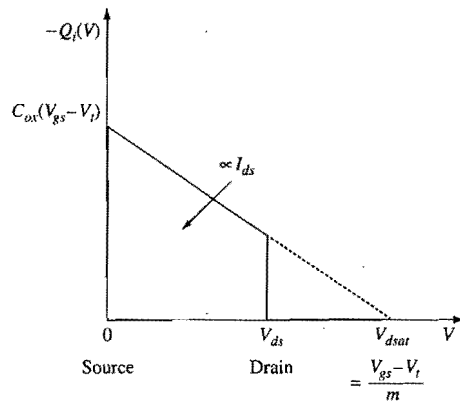


Figure 3.7. Inversion charge density as a function of the quasi-Fermi potential of a point in the channel. Before saturation, the drain current is proportional to the shaded area integrated from zero to the drain voltage.

where the simplified $Q_i(V)$, Eq. (3.29), has been used.

Substituting I_{ds} from Eq. (3.25) into Eq. (3.31), one can solve for $V(y)$:

$$V(y) = \frac{V_{gs} - V_t}{m} - \sqrt{\left(\frac{V_{gs} - V_t}{m}\right)^2 - 2\frac{y}{L}\left(\frac{V_{gs} - V_t}{m}\right)V_{ds} + \frac{y}{L}V_{ds}^2}. \quad (3.32)$$

Both $V(y)$ and $-Q_i/mC_{ox} = (V_{gs} - V_t)/m - V(y)$ are plotted in Fig. 3.9 for several values of V_{ds} . At low V_{ds} , $V(y)$ varies almost linearly between the source and drain. As V_{ds} increases, the inversion charge density at the drain decreases due to the lowering of the electron quasi-Fermi level. This is accompanied by a corresponding increase of dV/dy to maintain current continuity. When V_{ds} reaches $V_{dsat} = (V_{gs} - V_t)/m$, we have $Q_i(y=L) = 0$ and $V(y)$ exhibits a singularity at the drain, where $dV/dy = \infty$. This implies that **the electric field in the y-direction changes more rapidly than the field in the x-direction and the gradual channel approximation breaks down**. In other words, beyond the pinch-off point, carriers are no longer confined to the surface channel, and a 2-D Poisson equation must be solved for carrier injection from the pinch-off point into the drain depletion region (El-Mansy and Boothroyd, 1977).

Strictly speaking, if $V_{ds} > 2\psi_B$, Eq. (3.22) cannot be expanded into a power series in V_{ds} . A more general form of the saturation voltage is obtained by letting $Q_i = 0$ in Eq. (3.16) with $\psi_s = 2\psi_B + V$ and solving for $V = V_{dsat}$ [equivalent to solving $dI_{ds}/dV_{ds} = 0$ by differentiating Eq. (3.22)]:

$$V_{dsat} = V_{gs} - V_{fb} - 2\psi_B + \frac{\epsilon_{si}q N_a}{C_{ox}^2} - \sqrt{\frac{2\epsilon_{si}q N_a}{C_{ox}^2} \left(V_{gs} - V_{fb} + \frac{\epsilon_{si}q N_a}{2C_{ox}} \right)}. \quad (3.33)$$

The corresponding saturation current can be found by substituting Eq. (3.33) for V_{ds} in Eq. (3.22). The mathematics is rather tedious (Brews, 1981). A few selected curves are plotted in Fig. 3.10 and compared with those calculated from Eq. (3.25). It turns out that Eq. (3.25) serves as a good approximation to the drain current over a much wider range of

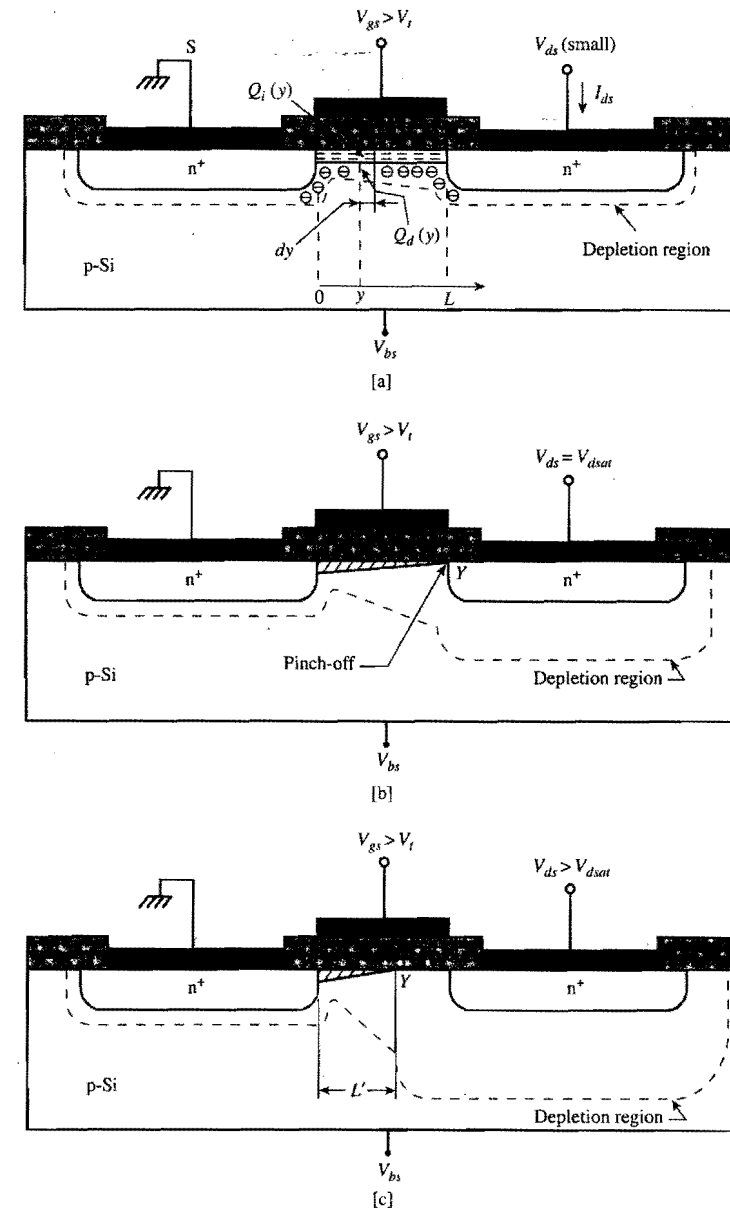


Figure 3.8. (a) MOSFET operated in the linear region (low drain voltage). (b) MOSFET operated at the onset of saturation. The pinch-off point is indicated by Y. (c) MOSFET operated beyond saturation where the channel length is reduced to L' . (After Sze, 1981.)

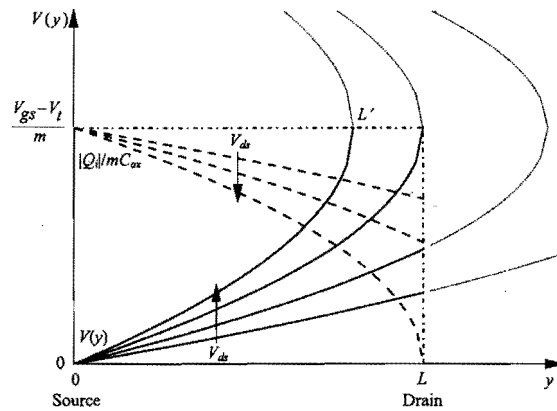


Figure 3.9. Quasi-Fermi potential versus distance between the source and the drain for several V_{ds} -values from the linear region to beyond saturation. The dashed curves show the corresponding variation of inversion charge density along the channel. The dotted curves help visualize the parabolic behavior of the characteristics.

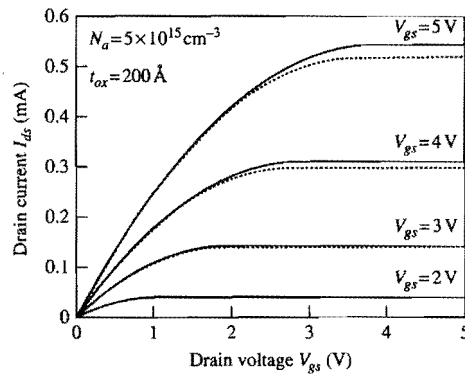


Figure 3.10. I_{ds} - V_{gs} curves calculated from the full equation (3.22) (solid curves), compared with the parabolic approximation (3.25) (dotted curves).

voltages than expected. Even for a drain voltage several times greater than $2\psi_B$, the current is only slightly ($\approx 5\%$) underestimated.

3.1.2.5 pMOSFET I - V Characteristics

So far we have used an n-channel device as an example to discuss MOSFET operation and I - V characteristics. A p-channel MOSFET operates similarly, except that it is

fabricated inside an n-well with implanted p^+ source and drain regions (cf. Fig. 3.2), and that the polarities of all the voltages and currents are reversed. For example, I_{ds} - V_{ds} characteristics for a pMOSFET (cf. Fig. 3.10) have negative gate and drain voltages with respect to the source terminal for a hole current to flow from the source to the drain.

Since the source of a pMOSFET is at the highest potential compared with the other terminals, it is usually connected to the power supply V_{dd} in a CMOS circuit so that all the voltages are positive (or zero). In that case, the device conducts if the gate voltage is lower than $V_{dd} - V_t$, where $V_t(>0)$ is the magnitude of the threshold voltage of the pMOSFET. The ohmic contact to the n-well is also connected to V_{dd} , in contrast to an nMOSFET, where the p-type substrate is usually tied to the ground potential. This leaves the n-well-to-p-substrate junction reverse biased. More about nMOSFET and pMOSFET bias conditions in a CMOS circuit configuration will be given in Section 5.1.

3.1.3 Subthreshold Characteristics

Depending on the gate and source-drain voltages, a MOSFET device can be biased in one of the three regions shown in Fig. 3.11. Linear (including parabolic) and saturation region characteristics have been described in the previous subsection. In this subsection, we discuss the characteristics of a MOSFET device in the subthreshold region where $V_{gs} < V_t$. In Fig. 3.4, the drain current on a linear scale appears to approach zero immediately below the threshold voltage. On a logarithmic scale, however, the descending drain current remains at nonnegligible levels for several tenths of a volt below V_t . This is because the inversion charge density does not drop to zero abruptly. Rather, it follows an exponential dependence on ψ_s or V_g , as is evident from Eq. (3.11). Subthreshold behavior is of particular importance in low-voltage, low-power applications, such as in digital logic and memory circuits, because it describes how a MOSFET device switches off. The subthreshold region immediately below V_t , in which $\psi_B \leq \psi_s \leq 2\psi_B$, is also called the *weak inversion region*.

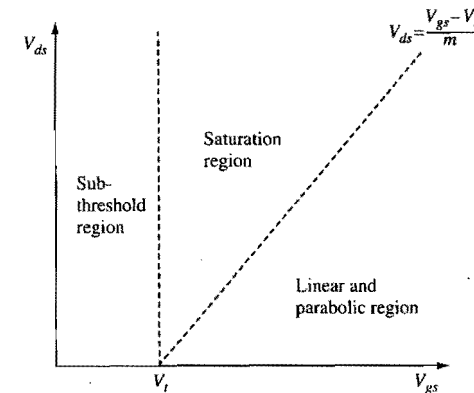


Figure 3.11. Three regions of MOSFET operation in the V_{ds} - V_{gs} plane.

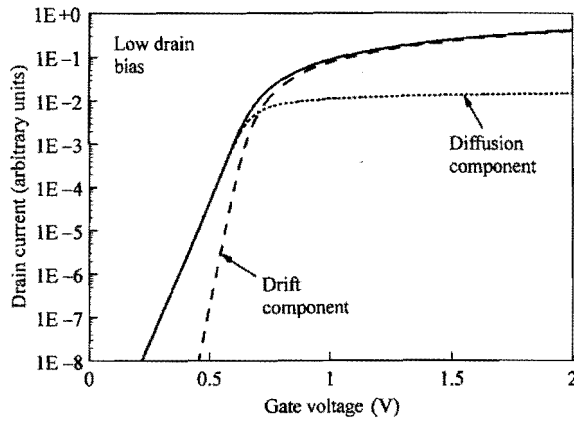


Figure 3.12. Drift and diffusion components of current in an I_{ds} - V_{gs} plot. Their sum is the total current represented by the solid curve.

3.1.3.1 Drift and Diffusion Components of Drain Current

Unlike the strong inversion region, in which the drift current dominates, subthreshold conduction is dominated by the diffusion current. Both current components are included in Pao and Sah's double integral, Eq. (3.13). In general, current continuity only applies to the total current, not to its individual components. In other words, the fractional ratio between the drift and the diffusion components may vary from one point of the channel to another. At low drain bias voltages, however, it is possible to separate the drift and diffusion components using the implicit $\psi_s(V)$ relation, Eq. (3.14). When $qV/kT \ll 1$, only the first-order terms of V need to be kept. In Eq. (3.8), $Q_i(V)$ can be replaced by its zeroth-order value, $Q_i(V=0)$; hence V must vary linearly from the source to the drain, as required by current continuity. Since the total current is proportional to dV/dy and the drift current is proportional to the electric field or $d\psi_s/dy$, the drift fraction of the current is given by the change of surface potential (band bending) with respect to the quasi-Fermi potential, i.e., $d\psi_s/dV$. This can be evaluated from Eq. (3.19) while making use of Eq. (3.14) in the limit of $V \rightarrow 0$:

$$\frac{d\psi_s}{dV} = \frac{(n_i^2/N_a^2)e^{q\psi_s/kT}}{1 + (n_i^2/N_a^2)e^{q\psi_s/kT} + (C_{ox}^2/\epsilon_{si}qN_a)(|Q_s|/C_{ox})}, \quad (3.34)$$

where $|Q_s|/C_{ox}$ is the voltage drop across the oxide given by the last term of Eq. (3.14). It is clear that in weak inversion where $\psi_B < \psi_s < 2\psi_B$, the numerator is much less than unity and the diffusion component dominates. Conversely, beyond strong inversion, $d\psi_s/dV \approx 1$ and the drift current dominates. These kinds of behavior are further illustrated in Fig. 3.12.

3.1.3.2 Subthreshold Current Expression

To find an expression for the subthreshold current, we note from Eq. (3.14) that the total charge density in silicon is,

$$-Q_s = \epsilon_{si} \mathcal{E}_s = \sqrt{2\epsilon_{si}kTN_a} \left[\frac{q\psi_s}{kT} + \frac{n_i^2}{N_a^2} e^{q(\psi_s - V)/kT} \right]^{1/2}. \quad (3.35)$$

In weak inversion, the second term in the brackets arising from the inversion charge density is much less than the first term from the depletion charge density. Equation (3.35) can then be expanded into a power series: the zeroth-order term is identified as the depletion charge density $-Q_d$ by Eq. (3.15), and the first-order term gives the inversion charge density,

$$-Q_i = \sqrt{\frac{\epsilon_{si}qN_a}{2\psi_s} \left(\frac{kT}{q} \right)} \left(\frac{n_i}{N_a} \right)^2 e^{q(\psi_s - V)/kT}. \quad (3.36)$$

The surface potential ψ_s is related to the gate voltage through Eq. (3.14). Since the inversion charge density is small, ψ_s is a function of V_{gs} only, independent of V (the case of $V_{gs} = 1$ V in Fig. 3.3). This also means that the electric field along the channel direction is small; hence the drift current is negligible.

Substituting Q_i into Eq. (3.10) and carrying out the integration, we obtain the drain current in the subthreshold region:

$$I_{ds} = \mu_{eff} \frac{W}{L} \sqrt{\frac{\epsilon_{si}qN_a}{2\psi_s} \left(\frac{kT}{q} \right)^2} \left(\frac{n_i}{N_a} \right)^2 e^{q\psi_s/kT} (1 - e^{-qV_{ds}/kT}). \quad (3.37)$$

ψ_s can be expressed in terms of V_{gs} using Eq. (3.14), where only the depletion charge term needs to be kept:

$$V_{gs} = V_{fb} + \psi_s + \frac{\sqrt{2\epsilon_{si}qN_a\psi_s}}{C_{ox}}. \quad (3.38)$$

It is straightforward to solve a quadratic equation for ψ_s . To further simplify the result, we consider ψ_s as only slightly deviated from the threshold value, $2\psi_B$ (Swanson and Meindl, 1972). Using the concept of $m = \Delta V_{gs}/\Delta\psi_s$ in Fig. 3.5, one can approximate V_{gs} as $V_{gs} = V_t + m(\psi_s - 2\psi_B)$. Solving for ψ_s and substituting it into Eq. (3.37) yield the subthreshold current as a function of V_{gs} :

$$I_{ds} = \mu_{eff} \frac{W}{L} \sqrt{\frac{\epsilon_{si}qN_a}{4\psi_B} \left(\frac{kT}{q} \right)^2} e^{q(V_{gs} - V_t)/mkT} (1 - e^{-qV_{ds}/kT}), \quad (3.39)$$

or

$$I_{ds} = \mu_{eff} C_{ox} \frac{W}{L} (m-1) \left(\frac{kT}{q} \right)^2 e^{q(V_{gs} - V_t)/mkT} (1 - e^{-qV_{ds}/kT}). \quad (3.40)$$

3.1.3.3 Subthreshold Slope

The subthreshold current is independent of the drain voltage once V_{ds} is larger than a few kT/q , as would be expected for diffusion-dominated current transport. The dependence on gate voltage, on the other hand, is exponential with an *inverse subthreshold slope* (Fig. 3.12),

$$S = \left(\frac{d(\log_{10} I_{ds})}{dV_{gs}} \right)^{-1} = 2.3 \frac{mkT}{q} = 2.3 \frac{kT}{q} \left(1 + \frac{C_{dm}}{C_{ox}} \right), \quad (3.41)$$

of typically 70–100 mV/decade. Here $m = 1 + (C_{dm}/C_{ox})$ from Eq. (3.27). If the Si–SiO₂ interface trap density is high, the subthreshold slope may be more graded than that given by Eq. (3.41), since the capacitance associated with the interface trap is in parallel with the depletion-layer capacitance C_{dm} . It should be noted that for ψ_s substantially below $2\psi_B$, e.g., when V_{gs} is a few tenths of a volt below V_t , Eq. (3.41) tends to underestimate the inverse subthreshold slope by 5–10%. As a result, the subthreshold current can be 2 to 4 times higher than that given by Eq. (3.40). For VLSI circuits, a steep subthreshold slope is desirable for the ease of switching the transistor current off. In MOSFET design, therefore, the gate oxide thickness and the bulk doping concentration should be chosen such that the factor m is not too much larger than unity, e.g., between 1.1 and 1.4. **Nevertheless, the inverse subthreshold slope has a lower bound of $2.3kT/q$, or 60 mV/decade at room temperature, that does not change with device parameters.** This has significant implications on device scaling as will be discussed in Chapter 4.

3.1.4 Substrate Bias and Temperature Dependence of Threshold Voltage

The threshold voltage is one of the key parameters of a MOSFET device. In this subsection, we examine the dependence of threshold voltage on substrate bias and temperature.

3.1.4.1 Substrate Sensitivity (Body Effect)

The drain-current equation in Section 3.1.2 was derived assuming zero substrate bias (V_{bs}). If $V_{bs} \neq 0$, one can modify the previously discussed MOSFET equations by considering that applying V_{bs} to the substrate is equivalent to subtracting all other voltages (namely, gate, source, and drain voltages) by V_{bs} while keeping the substrate grounded. This is shown in Fig. 3.13.

Using the charge-sheet model with $\psi_s = 2\psi_B + V$ as before, Eq. (3.16) becomes

$$Q_i = -C_{ox}(V_{gs} - V_{bs} - V_{fb} - 2\psi_B - V) + \sqrt{2\epsilon_{si}qN_a(2\psi_B + V)}, \quad (3.42)$$

where V is the reverse bias voltage between a point in the channel and the substrate.

The current is obtained by integrating Q_i from $V = -V_{bs}$ (source) to $V_{ds} - V_{bs}$ (drain):

$$I_{ds} = \mu_{eff} C_{ox} \frac{W}{L} \left\{ \left(V_{gs} - V_{fb} - 2\psi_B - \frac{V_{ds}}{2} \right) V_{ds} - \frac{2\sqrt{2\epsilon_{si}qN_a}}{3C_{ox}} \left[(2\psi_B - V_{bs} + V_{ds})^{3/2} - (2\psi_B - V_{bs})^{3/2} \right] \right\}. \quad (3.43)$$

At low drain voltages (linear region), the current is still given by Eq. (3.23), except that the threshold voltage is now

$$V_t = V_{fb} + 2\psi_B + \frac{\sqrt{2\epsilon_{si}qN_a(2\psi_B - V_{bs})}}{C_{ox}}. \quad (3.44)$$

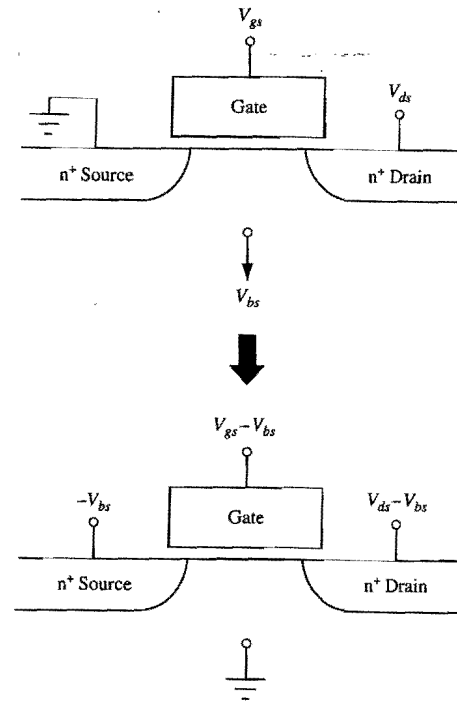


Figure 3.13. Equivalent circuits used to evaluate the effect of substrate bias on MOSFET I - V characteristics.

It can be seen from Eq. (3.44) that **the effect of a reverse substrate bias ($V_{bs} < 0$) is to widen the bulk depletion region and raise the threshold voltage.** Figure 3.14 plots V_t as a function of $-V_{bs}$. The slope of the curve,

$$\frac{dV_t}{d(-V_{bs})} = \frac{\sqrt{\epsilon_{si}qN_a/[2(2\psi_B - V_{bs})]}}{C_{ox}}, \quad (3.45)$$

is referred to as the *substrate sensitivity*. At $V_{bs} = 0$, the slope equals C_{dm}/C_{ox} , or $m - 1$ [Eq. (3.26)]. The substrate sensitivity is higher for a higher bulk doping concentration. It is clear from Fig. 3.14 that the substrate sensitivity decreases as the substrate reverse bias increases. From Eq. (3.41), a reverse substrate bias also makes the subthreshold slope slightly steeper, since it widens the depletion region and lowers C_{dm} .

3.1.4.2 Temperature Dependence of Threshold Voltage

Next, we examine the temperature dependence of the threshold voltage. The flat-band voltage of an nMOSFET with n⁺ polysilicon gate is $V_{fb} = -E_g/2q - \psi_B$ [Eq. (2.209)], assuming there is no oxide charge. Substituting it into Eq. (3.24) yields the threshold voltage,

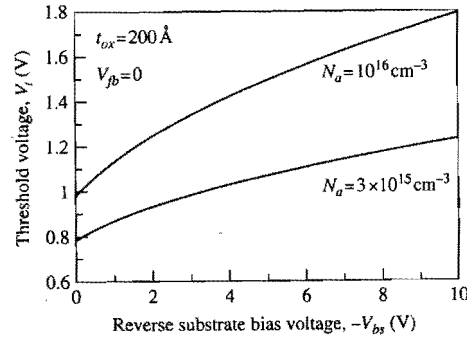


Figure 3.14. Threshold-voltage variation with reverse substrate bias for two uniform substrate doping concentrations.

$$V_t = -\frac{E_g}{2q} + \psi_B + \frac{\sqrt{4\epsilon_{si}qN_a\psi_B}}{C_{ox}}, \quad (3.46)$$

at zero substrate bias. The temperature dependence of V_t is related to the temperature dependence of E_g and ψ_B :

$$\begin{aligned} \frac{dV_t}{dT} &= -\frac{1}{2q} \frac{dE_g}{dT} + \left(1 + \frac{\sqrt{4\epsilon_{si}qN_a/\psi_B}}{C_{ox}}\right) \frac{d\psi_B}{dT} \\ &= -\frac{1}{2q} \frac{dE_g}{dT} + (2m-1) \frac{d\psi_B}{dT}. \end{aligned} \quad (3.47)$$

$d\psi_B/dT$ stems from the temperature dependence of the intrinsic carrier concentration, which can be evaluated using Eq. (2.48) and Eq. (2.13):

$$\begin{aligned} \frac{d\psi_B}{dT} &= \frac{d}{dT} \left[\frac{kT}{q} \ln \left(\frac{N_a}{\sqrt{N_c N_v} e^{-E_g/2kT}} \right) \right] \\ &= -\frac{k}{q} \ln \left(\frac{\sqrt{N_c N_v}}{N_a} \right) - \frac{kT}{q\sqrt{N_c N_v}} \frac{d\sqrt{N_c N_v}}{dT} + \frac{1}{2q} \frac{dE_g}{dT}. \end{aligned} \quad (3.48)$$

Since both N_c and N_v are proportional to $T^{3/2}$, we have $d(\sqrt{N_c N_v})/dT = \frac{3}{2}(\sqrt{N_c N_v})/T$. Substituting Eq. (3.48) into Eq. (3.47) yields

$$\frac{dV_t}{dT} = -(2m-1) \frac{k}{q} \left[\ln \left(\frac{\sqrt{N_c N_v}}{N_a} \right) + \frac{3}{2} \right] + \frac{m-1}{q} \frac{dE_g}{dT}. \quad (3.49)$$

From Section 2.1.1 and Table 2.1, $dE_g/dT \approx -2.7 \times 10^{-4}$ eV/K and $(N_c N_v)^{1/2} \approx 3 \times 10^{19}$ cm⁻³. For $N_a \sim 10^{16}$ cm⁻³ and $m \approx 1.1$, dV_t/dT is typically -1 mV/K. Note that the temperature coefficient decreases slightly as N_a increases: for $N_a \sim 10^{18}$ cm⁻³ and $m \approx 1.3$, dV_t/dT is about -0.7 mV/K. These numbers imply that, at an elevated temperature of, for example, 100°C, the threshold voltage is 55–75 mV lower than at room temperature. Since digital VLSI circuits often operate at elevated temperatures due to heat generation, this effect,

plus the degradation of subthreshold slope with temperature, causes the leakage current at $V_{gs} = 0$ to increase considerably over its room-temperature value. Typically, the off-state leakage current of a MOSFET at 100 °C is 30–50 times larger than the leakage current at 25 °C. These are important design considerations, to be addressed in detail in Chapter 4.

3.1.5 MOSFET Channel Mobility

The carrier mobility in a MOSFET channel is significantly lower than that in bulk silicon, due to additional scattering mechanisms. Lattice or phonon scattering is aggravated by the presence of crystalline discontinuity at the surface boundary, and surface roughness scattering severely degrades mobility at high normal fields. Channel mobility is also affected by processing conditions that alter the Si–SiO₂ interface properties (e.g., oxide charge and interface traps, as discussed in Section 2.3.6).

3.1.5.1 Effective Mobility and Effective Normal Field

In Section 3.1.1, the channel mobility was taken out of the integral by defining an effective mobility as

$$\mu_{eff} = \frac{\int_0^{x_i} \mu_n n(x) dx}{\int_0^{x_i} n(x) dx}, \quad (3.50)$$

which is essentially an average value weighted by the carrier concentration in the inversion layer. Empirically, it has been found that *when μ_{eff} is plotted against an effective normal field $\mathcal{E}_{eff}$, there exists a universal relationship independent of the substrate bias, doping concentration, and gate oxide thickness* (Sabnis and Clemens, 1979). The effective normal field is defined as the average electric field perpendicular to the Si–SiO₂ interface experienced by the carriers in the channel. Using Gauss's law, one can express $\mathcal{E}_{eff}$ in terms of the depletion and inversion charge densities:

$$\mathcal{E}_{eff} = \frac{1}{\epsilon_{si}} \left(|Q_d| + \frac{1}{2} |Q_i| \right), \quad (3.51)$$

where $|Q_d| + \frac{1}{2}|Q_i|$ is the total silicon charge inside a Gaussian surface through the middle of the inversion layer. Using Eq. (2.189) and Eq. (3.24), the depletion charge can be expressed as

$$|Q_d| = \sqrt{4\epsilon_{si}qN_a\psi_B} = C_{ox}(V_t - V_{fb} - 2\psi_B). \quad (3.52)$$

Substituting this expression and $|Q_i| \approx C_{ox}(V_{gs} - V_t)$ into Eq. (3.51) yields

$$\mathcal{E}_{eff} = \frac{V_t - V_{fb} - 2\psi_B}{3t_{ox}} + \frac{V_{gs} - V_t}{6t_{ox}}, \quad (3.53)$$

where $C_{ox} = \epsilon_{ox}/t_{ox}$ and $\epsilon_{si} \approx 3\epsilon_{ox}$ were used. Equation (3.53) can further be simplified if the gate electrode is n⁺ polysilicon (for nMOSFETs) such that $V_{fb} = -E_g/2q - \psi_B$. For submicron CMOS technologies, $\psi_B = 0.30$ – 0.42 V. Therefore, the effective normal field can be expressed in terms of explicit device parameters as

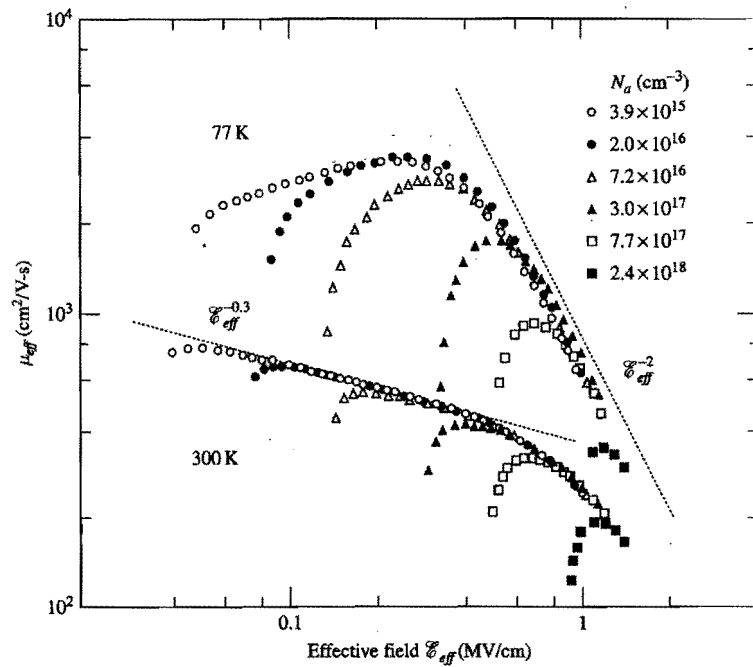


Figure 3.15. Measured electron mobility at 300 and 77 K versus effective normal field for several substrate doping concentrations. (After Takagi *et al.*, 1988).

$$\mathcal{E}_{eff} = \frac{V_t + 0.2}{3t_{ox}} + \frac{V_{gs} - V_t}{6t_{ox}}. \quad (3.54)$$

Equation (3.54) is valid for low drain voltages. At high drain voltages, Q_i decreases toward the drain end of the channel. To estimate the average effective field in that case, the second term in Eq. (3.54) should be reduced accordingly.

3.1.5.2 Electron Mobility Data

A typical set of data on mobility versus effective normal field for nMOSFETs is shown in Fig. 3.15 (Takagi *et al.*, 1988). At room temperature, the mobility follows a $\mathcal{E}_{eff}^{-1/3}$ dependence below 5×10^5 V/cm. A simple, approximate expression for this case is (Baccarani and Wordeman, 1983)

$$\mu_{eff} \approx 32500 \times \mathcal{E}_{eff}^{-1/3}. \quad (3.55)$$

Beyond $\mathcal{E}_{eff} = 5 \times 10^5$ V/cm, μ_{eff} decreases much more rapidly with increasing $\mathcal{E}_{eff}$ because of increased surface roughness scattering as carriers are distributed closer to the surface under high normal fields. For each doping concentration, there exists an effective field below which the mobility falls off the universal curve. This is believed to be due to Coulomb (or impurity) scattering, which becomes more important when the doping

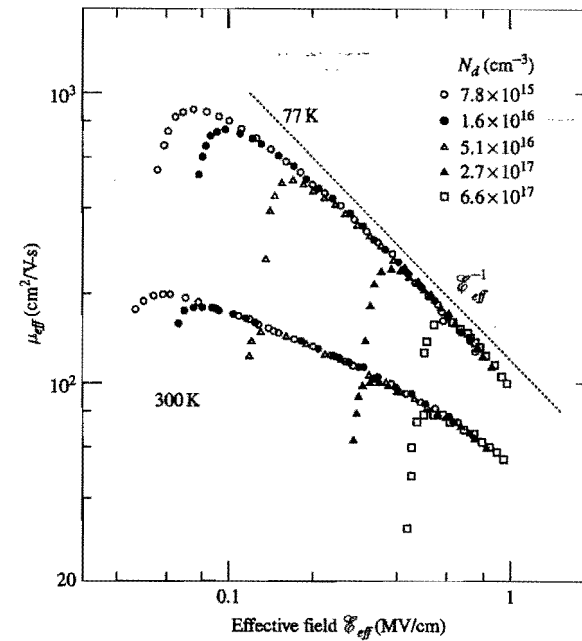


Figure 3.16. Measured hole mobility at 300 K and 77 K versus effective normal field (with a factor $\frac{1}{3}$) for several substrate doping concentrations. (After Takagi *et al.*, 1988).

concentration is high and the gate voltage or the normal field is low. There is less effect of Coulomb scattering on mobility when the inversion charge density is high because of charge screening effects. At 77 K, μ_{eff} is an even stronger function of $\mathcal{E}_{eff}$ and N_d . At low temperatures, surface scattering is the dominant mechanism at high fields, while Coulomb scattering dominates at low fields.

3.1.5.3 Hole Mobility Data

Similar mobility-field data for pMOSFETs are shown in Fig. 3.16. In this case, however, the effective normal field is defined by

$$\mathcal{E}_{eff} = \frac{1}{\epsilon_{si}} \left(|Q_d| + \frac{1}{3} |Q_i| \right), \quad (3.56)$$

which has been found necessary in order for the measured hole mobilities to fall on a universal curve when plotted against $\mathcal{E}_{eff}$ (Arora and Gildenblat, 1987). Note that the factor $\frac{1}{3}$ is entirely empirical with no physical reasoning behind it. Like electron mobility, hole mobility is also influenced by Coulomb scattering at low fields, depending on the doping concentration. The field dependence is also stronger at 77 K, but not quite as strong as in the electron case. It should be noted that the hole mobility data were taken from surface-channel pMOSFETs with p⁺ polysilicon gate. Buried-channel pMOSFETs with n⁺ polysilicon gate have a higher mobility (by about 30%) for the same threshold

voltage. This is because the normal field in a buried-channel device, given by Eq. (3.53) with $V_{fb} = +E_g/2q - \psi_B$ (p^+ gate on nMOSFET as an analogy), is much lower. However, buried-channel MOSFETs cannot be scaled to as short a channel length as surface-channel MOSFETs, as will be discussed in Chapter 4.

At higher temperatures, the MOSFET channel mobility decreases because of increased phonon scattering. The temperature dependence is similar to that of bulk mobility discussed in Section 2.1.3, i.e., $\mu_{eff} \propto T^{-3/2}$.

3.1.6 MOSFET Capacitances and Inversion-Layer Capacitance Effect

In this subsection, we discuss the intrinsic capacitances of a MOSFET device in different regions of operation and the effect of finite inversion-layer capacitance on linear $I_{ds} - V_{gs}$ characteristics, which has been neglected in the regional model.

3.1.6.1 Intrinsic MOSFET Capacitances

The capacitance of a MOSFET device plays a key role in the switching delay of a logic gate, since for a given current, the capacitance determines how fast the gate can be charged (or discharged) to a certain potential which turns on (or off) the source-to-drain current. MOSFET capacitances can be divided into two main categories: intrinsic capacitances and parasitic capacitances. This sub-subsection focuses on the intrinsic MOSFET capacitances arising from the inversion and depletion charges in the channel region. Parasitic capacitances are discussed in Section 5.2. As in the earlier drain-current discussions, gate capacitances are also considered separately in the three regions of MOSFET operation: subthreshold region, linear region, and saturation region, as shown in Fig. 3.11.

- **Subthreshold region.** In the subthreshold region, the inversion charge is negligible. Only the depletion charge needs to be supplied when the gate potential is changed. Therefore, the intrinsic gate-to-source-drain capacitance is essentially zero (the extrinsic gate-to-source-drain overlap capacitance is discussed in Section 5.2.2), while the gate-to-body capacitance is given by the serial combination of C_{ox} and C_d (Fig. 2.37), i.e.,

$$C_G = WL \left(\frac{1}{C_{ox}} + \frac{1}{C_d} \right)^{-1} \approx WLC_d, \quad (3.57)$$

where C_d is the depletion capacitance per unit area given by Eq. (2.201). Here, the upper-case subscript is used to distinguish the total gate capacitance C_G from the gate capacitance per unit area, C_g . For high drain biases, the surface potential and therefore the depletion width at the drain end of the channel become larger, according to Eq. (3.4). The average C_d to be used in Eq. (3.57) should then be slightly lower than that evaluated at the source end.

- **Linear region.** Once the surface channel forms, there is no more capacitive coupling between the gate and the body due to screening by the inversion charge. All the gate capacitances are to the channel, i.e., to the source and drain terminals. Within the

framework of the regional charge-sheet model, Eq. (3.29), the inversion charge density Q_i at low drain biases varies linearly from $-C_{ox}(V_{gs} - V_t)$ at the source end to $-C_{ox}(V_{gs} - V_t - mV_{ds})$ at the drain end. The total inversion charge under the gate is then $-WLC_{ox}(V_{gs} - V_t - mV_{ds}/2)$, and the gate to channel capacitance is simply given by the oxide capacitance,

$$C_G = WLC_{ox}. \quad (3.58)$$

- **Saturation region.** When V_{ds} is appreciable, the inversion charge density $Q_i(y)$ varies parabolically along the channel, as shown in Fig. 3.9. At the pinch-off condition, $V_{ds} = V_{dsat} = (V_{gs} - V_t)/m$, and $Q_i = 0$ at the drain. In this case,

$$Q_i(y) = -C_{ox}(V_{gs} - V_t) \sqrt{1 - \frac{y}{L}}, \quad (3.59)$$

from Eqs. (3.29) and (3.32). The total inversion charge obtained by integrating Eq. (3.59) in both the channel length (y) and the channel width directions is then $-\frac{2}{3}WLC_{ox}(V_{gs} - V_t)$, and the gate-to-channel capacitance in the saturation region is

$$C_G = \frac{2}{3}WLC_{ox}. \quad (3.60)$$

3.1.6.2 Inversion-Layer Capacitance

In Section 3.1.2 and in the discussions above, MOSFET $I-V$ relations and capacitances were derived based on the regional approximation that all the inversion charge is located at the silicon surface and the surface potential is pinned at $\psi_s(\text{inv}) = 2\psi_B$ once the inversion layer forms. In reality, the inversion layer has a finite thickness (Fig. 2.34), and the surface potential still increases slightly with V_{gs} even beyond $2\psi_B$ (Fig. 2.36). In other words, there is a finite inversion-layer capacitance, $C_i = -dQ_i/d\psi_s$, in series with the oxide capacitance. As a result, the inversion charge density is less than that given by Eq. (3.29). The error is illustrated in the $Q_i - V_{gs}$ curves in Fig. 3.17. The dashed line represents $|Q_i| = C_{ox}(V_{gs} - V_t)$ from the piecewise model. The solid curve is a more exact solution calculated numerically from Eq. (3.12) and Eq. (3.14). The discrepancy extends to high gate voltages but is more serious for low-voltage operation. An approximate expression for the inversion charge density taking this effect into account can be derived by considering the small-signal capacitances in Fig. 2.37(b) (Wordeman, 1986),

$$\frac{d(-Q_i)}{dV_{gs}} = \frac{C_{ox}C_i}{C_{ox} + C_i + C_d} \approx C_{ox} \left(1 - \frac{1}{1 + C_i/C_{ox}} \right). \quad (3.61)$$

Here $C_d \approx 0$ after the onset of strong inversion because of screening by the inversion charge. From Eq. (2.206), $C_i \approx |Q_i|/(2kT/q)$. Since $|Q_i| \approx C_{ox}(V_{gs} - V_t)$, one can write $C_i/C_{ox} = (V_{gs} - V_t)/(2kT/q)$. Substituting it into Eq. (3.61) and integrating with respect to V_{gs} , one obtains

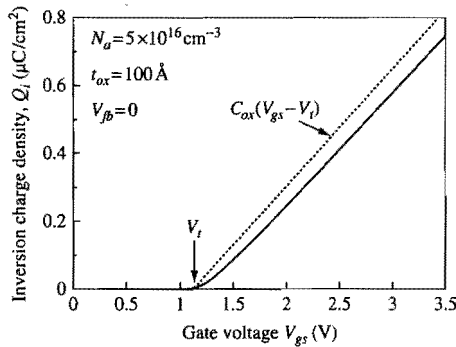


Figure 3.17. $Q_i - V_{gs}$ curve (solid line) calculated from Pao and Sah's model for zero drain voltage, compared with that of the regional approximation (dotted line). V_t indicates the $2\psi_B$ threshold.

$$-Q_i = C_{ox} \left[(V_{gs} - V_t) - \frac{2kT}{q} \ln \left(1 + \frac{q(V_{gs} - V_t)}{2kT} \right) \right], \quad (3.62)$$

which agrees well with the numerically calculated curve in Fig. 3.17.

3.1.6.3 Effect of Polysilicon-Gate Depletion on Inversion Charge

Depletion of polysilicon gates, discussed in Section 2.3.4, can also have an effect on the $Q_i - V_{gs}$ curve if the gate is not doped highly enough. To first order, the depletion region in polysilicon acts like a large capacitor in series with the oxide capacitor, which further degrades inversion charge density for a given applied gate voltage. In contrast to the inversion-layer capacitance effect, however, the gate depletion effect becomes more severe at high gate voltages. Assuming an n^+ polysilicon gate on nMOSFET (and vice versa for pMOSFET) and following a similar approach to that in Eq. (2.213), one can add an additional term to Eq. (3.62) for polysilicon depletion effects:

$$-Q_i = C_{ox} \left[(V_{gs} - V_t) - \frac{2kT}{q} \ln \left(1 + \frac{q(V_{gs} - V_t)}{2kT} \right) - \frac{C_{ox}^2 (V_{gs} - V_t)^2}{2\epsilon_{si} q N_p} \right]. \quad (3.63)$$

Here N_p is the electrically active doping concentration of the polysilicon gate, and the gate charge density Q_g has been approximated by $C_{ox}(V_{gs} - V_t)$ (ignoring the depletion charge in bulk silicon). Note that the factor of $\frac{1}{2}$ in the polysilicon depletion term arises from integrating a gate-voltage-dependent capacitance with respect to the gate voltage. In order to keep the last degradation term negligible, N_p should be in the range of 10^{20} cm^{-3} , especially for thin-oxide MOSFETs.

3.1.6.4 Linear $I_{ds} - V_{gs}$ Characteristics

Given $Q_i(V_{gs})$ and $\mu_{eff}(V_{gs})$ [Eqs. (3.55) and (3.54)], the low-drain-bias (linear) $I_{ds} - V_{gs}$ curve is simply

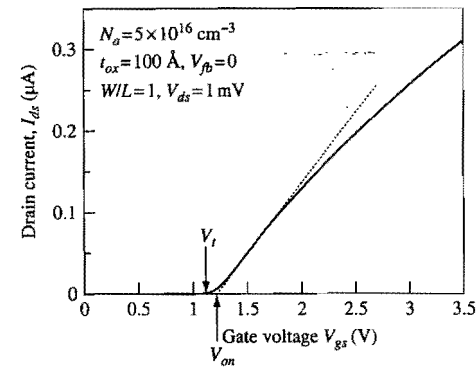


Figure 3.18. Calculated low-drain $I_{ds} - V_{gs}$ curve with inversion-layer capacitance and mobility degradation effects. The dotted line shows the linearly extrapolated threshold voltage V_{on} .

$$I_{ds}(V_{gs}) = \mu_{eff}(V_{gs}) \frac{W}{L} [-Q_i(V_{gs})] V_{ds}. \quad (3.64)$$

An example is shown in Fig. 3.18, where I_{ds} is calculated assuming no polysilicon depletion with $Q_i(V_{gs})$ from Fig. 3.17 (solid curve). Both the drain current and the linear transconductance, defined by $g_m \equiv dI_{ds}/dV_{gs}$, are degraded significantly at high gate voltages because of the decrease of mobility with increasing normal field. There is a point of maximum slope or linear transconductance about 0.5 V above the threshold voltage. It is conventional to define the linearly extrapolated threshold voltage, V_{on} , by the intercept of a tangent through this point. For a second-order correction in V_{ds} based on Eq. (3.25), V_{on} is obtained by subtracting $mV_{ds}/2$ from the intercept. Because of the combined inversion-layer capacitance and mobility degradation effects, the linearly extrapolated threshold voltage, V_{on} , is typically $(2-4)kT/q$ higher than the threshold voltage V_t at $\psi_s(\text{inv}) = 2\psi_B$. One should be careful not to mix up V_{on} with V_t , which is used in Eq. (3.40) for estimating subthreshold currents. At $V_{gs} = V_{on}$, the extrapolated subthreshold current (along the same subthreshold slope in a semilog plot) is about $10\times$ of that at $V_{gs} = V_t$. This current is rather insensitive to temperature but does depend on the technology generation.

The inversion layer capacitance in Eq. (3.62) was derived assuming classical density of states with Boltzmann statistics. The inversion layer is actually deeper than the classical value due to quantum effects (Section 4.2.4) which further degrade Q_i from that of Eq. (3.63). It is a common practice to lump all these effects into a parameter called t_{inv} by defining $-dQ_i/dV_{gs} = C_{inv} = \epsilon_{ox}/t_{inv}$. In general, t_{inv} is a function of V_{gs} and is 5–10 Å thicker than the physical t_{ox} . A split $C-V$ measurement like that described in Fig. 2.39 is needed to separate the t_{inv} -factor from $\mu_{eff}(V_{gs})$ in Eq. (3.64).

3.2

Short-Channel MOSFETs

It is clear from Section 3.1 that for a given supply voltage, the MOSFET current increases with decreasing channel length. The intrinsic capacitance of a short-channel MOSFET is

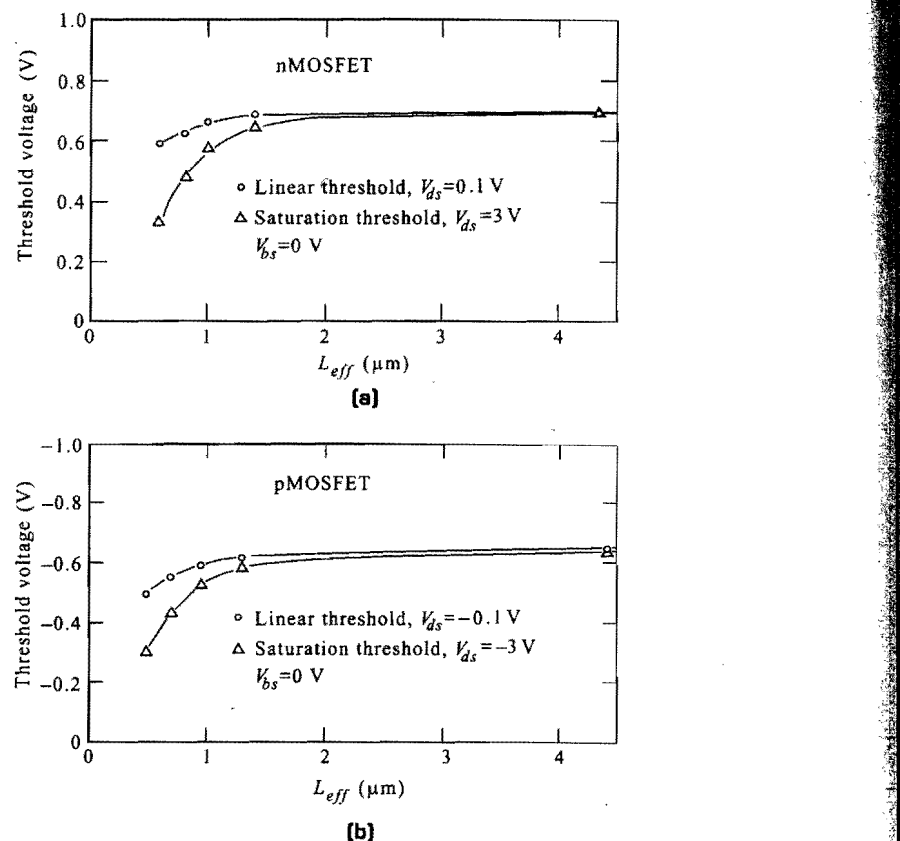


Figure 3.19. Short-channel threshold rolloff: Measured low- and high-drain threshold voltages of n- and p-MOSFETs versus channel length. (After Taur *et al.*, 1985.)

also lower, which makes it easier to switch. However, for a given process, the channel length cannot be arbitrarily reduced even if allowed by lithography. Short-channel MOSFETs differ in many important aspects from long-channel devices discussed in Section 3.1. This section covers the basic features of short-channel devices that are important for device design consideration. These features are: (a) short-channel effect, (b) velocity saturation, (c) channel length modulation, (d) source-drain series resistance, and (e) MOSFET degradation and breakdown.

3.2.1 Short-Channel Effect

The *short-channel effect* (SCE) is the decrease of the MOSFET threshold voltage as the channel length is reduced. An example is shown in Fig. 3.19 (Taur *et al.*, 1985). The short-channel effect is especially pronounced when the drain is biased at a voltage equal

to that of the power supply (high drain bias). In a CMOS VLSI technology, channel length varies statistically from chip to chip, wafer to wafer, and lot to lot due to process tolerances. The short-channel effect is therefore an important consideration in device design; one must ensure that the threshold voltage does not become too low for the minimum-channel-length device on the chip.

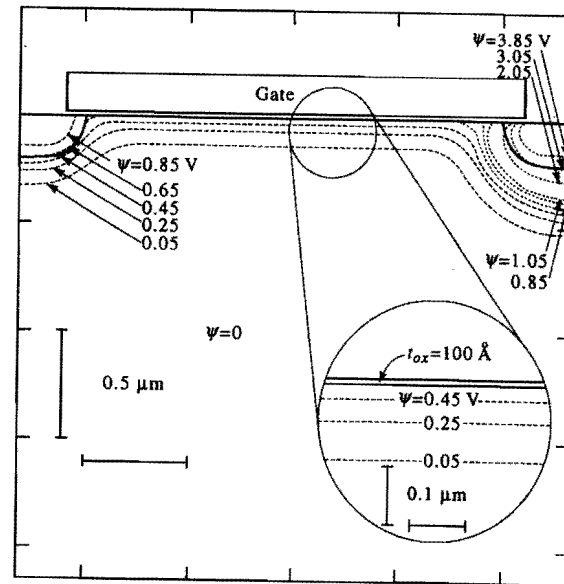
3.2.1.1 2-D Potential Contours in a MOSFET

The key difference between a short-channel and a long-channel MOSFET is that the field pattern in the depletion region of a short-channel MOSFET is two-dimensional, as shown in Fig. 3.20. The constant-potential contours in a long-channel device in Fig. 3.20(a) are largely parallel to the oxide-silicon interface or along the channel length direction (y -axis), so that the electric field is one-dimensional (along the vertical direction or x -axis) over the most part of the device. The constant-potential contours in a short-channel device in Fig. 3.20(b), however, are more curvilinear, and the resulting electric field pattern is of a two-dimensional nature. In other words, both the x - and y -components of the electric field are appreciable in a short-channel MOSFET. It is also important to note that, for a given gate voltage, there is more band bending (higher ψ) at the silicon-oxide interface in a short-channel device than in a long-channel device. Specifically, the maximum surface potential is slightly over 0.65 V (the fourth contour from the bottom) in Fig. 3.20(b), but below 0.65 V in Fig. 3.20(a). The depletion region width, as indicated by the depth of the first contour ($\psi = 0.05\text{ V}$) from the bottom, is also wider in the short-channel case. These all point to a lower threshold voltage in the short-channel MOSFET.

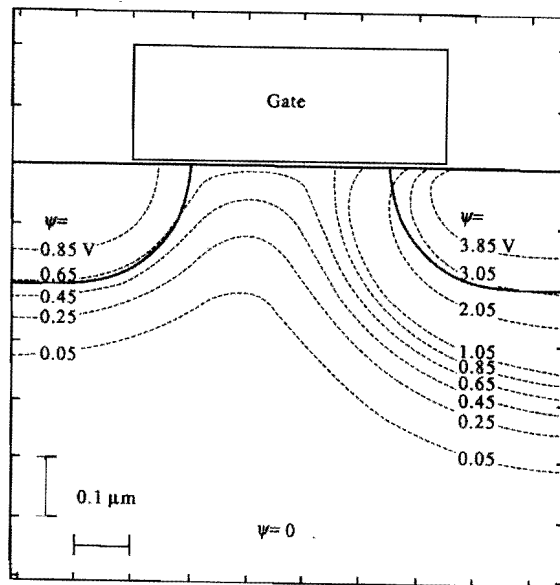
The two-dimensional field pattern in a short-channel device arises from the proximity of source and drain regions. Just like the depletion region under an MOS gate (Section 2.3), there are also depletion regions surrounding the source and drain junctions (Section 2.2 and Fig. 2.44). In a long-channel device, the source and drain are far enough separated that their depletion regions have no effect on the potential or field pattern in most parts of the device. *In a short-channel device, however, the source-drain distance is comparable to the MOS depletion width in the vertical direction, and the source-drain potential has a strong effect on the band bending over a significant portion of the device.*

3.2.1.2 Drain-Induced Barrier Lowering

The physics of the short-channel effect can be understood from another angle by considering the potential barrier (to electrons for an n-channel MOSFET) at the surface between the source and drain, as shown in Fig. 3.21 (Troutman, 1979). Under off conditions, this potential barrier (p-type region) prevents electron current from flowing to the drain. The surface potential is mainly controlled by the gate voltage. When the gate voltage is below the threshold voltage, there are only a limited number of electrons injected from the source over the barrier and collected by the drain (subthreshold current). In the long-channel case, the potential barrier is flat over most parts of the device. Source and drain fields only affect the very ends of the channel. As the channel length is shortened, however, the source and drain fields penetrate deeply into the middle of the channel, which lowers the potential barrier between the source and drain. This



(a)



(b)

Figure 3.20. Simulated constant potential contours of (a) a long-channel and (b) a short-channel nMOSFET. The contours are labeled by the band bending with respect to the neutral p-type region. The solid lines indicate the location of the source and drain junctions (metallurgical). The drain is biased at 3.0 V. Both devices are biased at the same gate voltage slightly below the threshold.

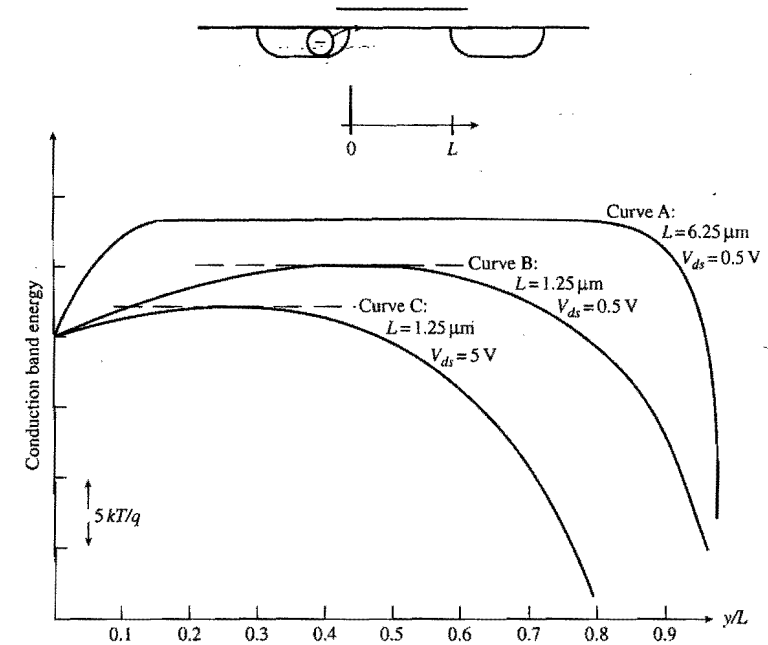


Figure 3.21. Conduction band energy versus lateral distance (normalized to the channel length L) from the source to the drain for (A) a long-channel MOSFET, (B) a short-channel MOSFET at low drain bias, and (C) a short-channel MOSFET at high drain bias. The gate voltage is the same for all three cases. (After Troutman, 1979.)

causes a substantial increase of the subthreshold current. In other words, the threshold voltage becomes lower than the long-channel value. The region of maximum potential barrier also shrinks to a single point near the center of the device.

When a high drain voltage is applied to a short-channel device, the barrier height is lowered even more, resulting in further decrease of the threshold voltage. The point of maximum barrier also shifts toward the source end as shown in Fig. 3.21. This effect is referred to as *drain-induced barrier lowering (DIBL)*. It explains the experimentally observed increase of subthreshold current with drain voltage in a short-channel MOSFET. Figure 3.22 shows the subthreshold characteristics of long- and short-channel devices at different drain bias voltages. For long-channel devices, the subthreshold current is independent of drain voltage ($\geq 2kT/q$), as expected from Eq. (3.40). For short-channel devices, however, there is a parallel shift of the curve to a lower threshold voltage under high drain bias conditions. At even shorter channel lengths, the subthreshold slope starts to degrade as the surface potential is more controlled by the drain than by the gate. Eventually, the device reaches the *punch-through* condition when the gate totally loses control of the channel and high drain current persists independent of gate voltage.

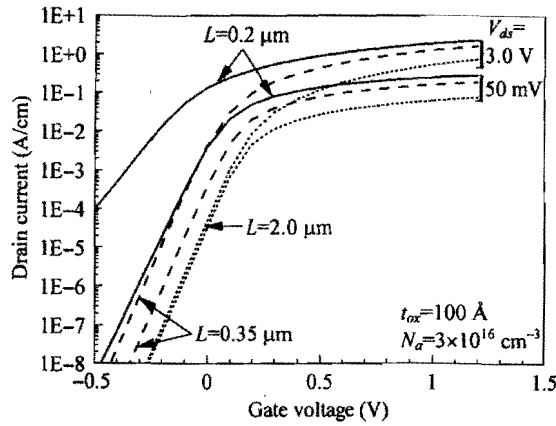


Figure 3.22. Subthreshold characteristics of long- and short-channel devices at low and high drain bias.

3.2.1.3 2-D Poisson Equation and Lateral Field Penetration

Further insight into the role of the lateral electric field, $\mathcal{E}_y = -\partial\psi_i/\partial y$, in a short-channel MOSFET can be gained by examining the two-dimensional Poisson equation,

$$\frac{\partial^2 \psi_i}{\partial x^2} + \frac{\partial^2 \psi_i}{\partial y^2} = -\frac{\rho}{\epsilon_{si}}. \quad (3.65)$$

In the depletion region of an nMOSFET, mobile carrier densities are negligible. Only ionized acceptors need to be considered. For a uniformly doped background concentration N_a , Poisson's equation can be written in terms of the electric fields as

$$\frac{\partial \mathcal{E}_x}{\partial x} + \frac{\partial \mathcal{E}_y}{\partial y} = \frac{\rho}{\epsilon_{si}} = -\frac{qN_a}{\epsilon_{si}}, \quad (3.66)$$

where $\mathcal{E}_x = -\partial\psi_i/\partial x$ is the electric field in the vertical direction. The depletion charge density, $\rho = -qN_a$, from ionized acceptors can be considered as being split into two parts: the first part, $\epsilon_{si}\partial\mathcal{E}_x/\partial x$, is controlled by the gate field in the vertical direction; the second part, $\epsilon_{si}\partial\mathcal{E}_y/\partial y$, is controlled by the source-drain field in the lateral direction (Nguyen and Plummer, 1981). In a long-channel device, the lateral field is negligible over most of the channel, and almost all of the depletion charge is controlled by the gate field. In a short-channel device, the lateral field becomes appreciable. Figure 3.23(a) shows an example of the magnitude of the lateral field along the channel length direction obtained from a 2-D numerical simulation. The lateral field is highest at the source and drain junctions, decreasing exponentially toward the middle of the channel. At low drain voltages, the source and drain fields cancel each other exactly at the center of the device.

When the channel length becomes shorter, the characteristic length of the exponential decay remains unchanged, while the magnitude of the lateral field near the middle of the device increases significantly. This depicts the penetration of source and drain fields into the channel region of a short-channel MOSFET. It is shown in Appendix 9 that the

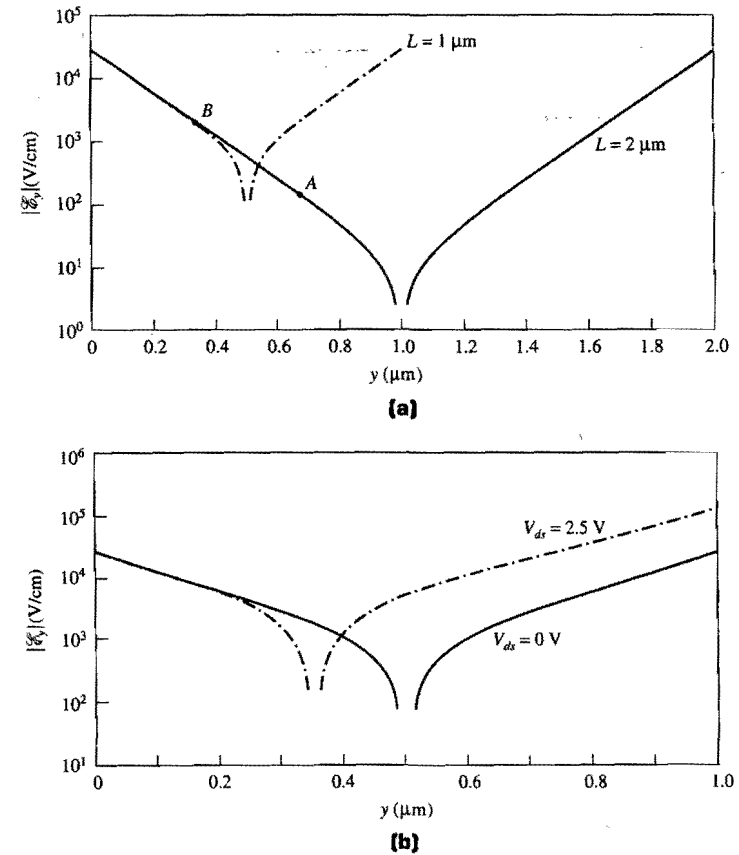


Figure 3.23. Simulated lateral field as a function of lateral distance along a horizontal cut through the gate-depletion layer for (a) long- and short-channel devices and (b) low and high drain bias voltages. (After Nguyen, 1984.)

characteristic length of the exponential decay of lateral fields is $(W_d + 3t_{ox})/\pi$ [Eq. (A9.22)], where W_d is the depth of the gate depletion region. Application of a high drain voltage [Fig. 3.23(b)] does not change the source field but does increase the drain field. This shifts the zero-field point toward the source, thus making it asymmetric, and at the same time raises the lateral field intensity even further. The zero-field point corresponds to the point of the shallowest depletion depth in Fig. 3.20(b), as well as the point of maximum energy barrier in Fig. 3.21.

3.2.1.4 An Analytical Expression for Short-Channel Threshold Voltage

With a few approximations, an analytical solution to the two-dimensional Poisson equation can be obtained using a simplified short-channel MOSFET geometry in Appendix 9 (Nguyen, 1984). The region of interest is a rectangular box of length equal

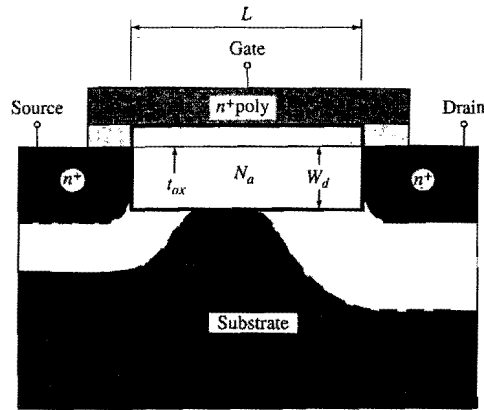


Figure 3.24. Simplified geometry for analytically solving Poisson's equation in a short-channel MOSFET. (After Nguyen, 1984.)

to the channel length L defined as the distance between the source and the drain (Fig. 3.24). In the vertical direction, the box consists of an oxide region of thickness t_{ox} and a silicon region of depth given by the depletion-layer width W_d [Eq. (2.188)]. To eliminate the discontinuity of $\partial\psi/\partial x$ across the silicon-oxide boundary, the oxide is replaced by an equivalent region of the same dielectric constant as silicon, but with a thickness equal to $(\epsilon_{si}/\epsilon_{ox})t_{ox} = 3t_{ox}$. The entire rectangular region can then be treated as a homogeneous material of height $W_d + 3t_{ox}$ and dielectric constant ϵ_{si} . This is a good approximation when the oxide is thin compared with the depletion depth W_d , as is the case with most practical CMOS technologies.

The boundary conditions of the electrostatic potential at the source and drain boundaries are ψ_{bi} and $\psi_{bi} + V_{ds}$, respectively, with the potential in the neutral p-type region defined as zero. Here ψ_{bi} is the built-in potential of the source- or drain-to-substrate junction, and V_{ds} is the drain voltage. For an abrupt n^+ -p junction, $\psi_{bi} = E_g/2q + \psi_B$, where ψ_B is given by Eq. (2.48). Typically, $\psi_{bi} \approx 0.8$ – 0.9 V.

Under subthreshold conditions, current conduction is dominated by diffusion and is mainly controlled by the point of highest barrier for electrons along the channel, as shown in Fig. 3.20(b) and Fig. 3.21. The threshold voltage of a short-channel device is defined as the gate voltage at which the minimum electrostatic potential (maximum barrier for electrons) at the surface equals $2\psi_B$. It is shown in Appendix 9 [Eq. (A9.25)] that this occurs at a gate voltage lower than the long-channel threshold voltage by an amount

$$\Delta V_t = \frac{24t_{ox}}{W_{dm}} \left[\sqrt{\psi_{bi}(\psi_{bi} + V_{ds})} - a(2\psi_B) \right] e^{-\pi L/2} \quad (3.67)$$

Here $a \approx 0.4$ and W_{dm} is the maximum depletion width at the threshold condition, $\psi_s = 2\psi_B$. For typical values of $\psi_{bi} \approx V_{ds} \approx 2\psi_B \approx 1$ V, and $3t_{ox}/W_{dm} = m - 1 \approx 0.3$, Eq. (3.67) gives a short-channel V_t rolloff of 100 mV for $L = 2(W_{dm} + 3t_{ox})$. Since 100 mV is the

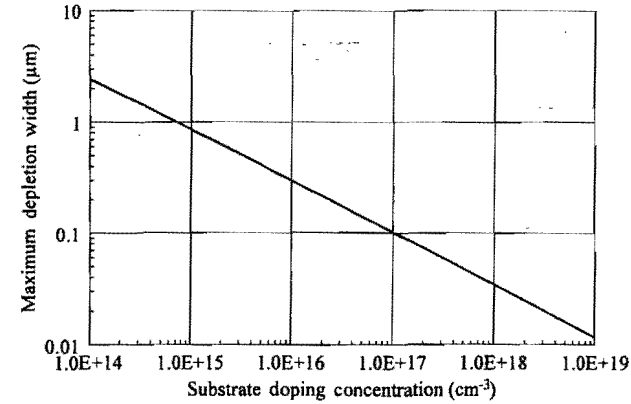


Figure 3.25. Depletion region width at $2\psi_B$ threshold condition versus doping concentration for uniformly doped substrates.

generally accepted worst-case V_t rolloff in a modern CMOS technology, **the minimum allowable channel length is $L_{min} \approx 2(W_{dm} + 3t_{ox})$** . Note that $W_{dm} + 3t_{ox}$ is the effective height of the rectangular box in Fig. 3.24, and L is its width. Qualitatively, the severity of the short-channel effect is measured by the aspect ratio of the rectangular box. A low aspect ratio such that $L/(W_{dm} + 3t_{ox}) > 2$ assures acceptable short-channel effects.

Because of the exponential factor, the threshold voltage rolloff with channel length is very sensitive to $W_{dm} + 3t_{ox}$, which can be defined as the scale length λ of the MOSFET. The criterion for the minimum channel length is then $L_{min} \approx 2\lambda$. To scale a MOSFET to shorter channel lengths with acceptable short-channel effects, the scale length λ needs to be reduced accordingly. This means scaling down both W_{dm} and t_{ox} by the same factor as the channel length. Note that for a uniformly doped substrate,

$$W_{dm} = \sqrt{\frac{4\epsilon_{si}kT \ln(N_a/n_i)}{q^2 N_a}} \quad (3.68)$$

from Eq. (2.190). W_{dm} is plotted in Fig. 3.25 versus N_a .

The above W_{dm} is that of a long-channel device, independent of L . This is a good approximation for $L \geq 2(W_{dm} + 3t_{ox})$. For shorter channel lengths, W_{dm} tends to increase as L decreases (see Fig. 3.20). When that happens, ΔV_t does not increase as rapidly with decreasing L as indicated by the exponential factor in Eq. (3.67). For $L \approx 1.5(W_{dm} + 3t_{ox})$, Eq. (3.67) with the long-channel W_{dm} tends to over estimate the threshold rolloff by a factor of ≈ 1.3 (Kannan, 2005). In aggressively scaled, high-performance CMOS logic technologies, L_{min} is often pushed to $\approx 1.5(W_{dm} + 3t_{ox})$. For $L \leq W_{dm} + 3t_{ox}$, the assumption that higher order terms in u_L and u_R series (Section A9.2) are negligible is no longer valid. Such devices have too severe a short-channel effect to be of practical use anyway.

All of the above discussions assume that the source and drain junction depth, x_j , is larger than the depletion region width, W_{dm} . It led to the result that V_t rolloff is controlled

by W_{dm} , insensitive to the junction depth. This is also the technologically relevant case since in practice it is difficult to scale down the junction depth without degrading the device current due to increased series resistance. But if a MOSFET with $x_j < W_{dm}$ can be made (e.g., by raised source-drain process), V_t rolloff will be linearly improved in proportion to x_j/W_{dm} (Sleva and Taur, 2005).

While analytical results like Eq. (3.67) give us key insights to the short-channel effect, in general short-channel device design is carried out with a two-dimensional device simulator for more accurate results. Further details on channel profile and threshold design are discussed in Section 4.2.

3.2.1.5 Generalized Scale Length with High- κ Gate Dielectrics

When the CMOS channel length is scaled to 20–30 nm, gate oxides of ≈ 1 nm thickness (Section 4.2.3) become necessary for control of short-channel effects. While a combination of improvement in process technology and better understanding of the breakdown process of gate oxides (Section 2.5.6) has made this possible, gate tunneling currents can be unacceptably high for such atomically thin oxides (Fig. 2.62). This problem can be mitigated using high-permittivity (high- κ) dielectrics to replace SiO_2 as gate insulators. From the normal field (in silicon) and gate capacitance point of view, a high- κ gate dielectric of permittivity ϵ_i and thickness

$$t_i = \frac{\epsilon_i}{\epsilon_{ox}} t_{ox} \quad (3.69)$$

is equivalent to an oxide layer of permittivity ϵ_{ox} and thickness t_{ox} . If $\epsilon_i/\epsilon_{ox} \gg 1$, the physical thickness of the high- κ gate dielectric t_i is much thicker than t_{ox} , thus significantly reducing the gate tunneling current (quantum mechanical tunneling has nothing to do with the dielectric constant of the material). In practice, it is rather difficult to develop a high- κ gate insulator with acceptable characteristics for use in CMOS products. High- κ gate insulator is currently one of the intensely researched subjects in the field of VLSI.

From the one-region scale length model, one would expect that $\lambda = W_{dm} + (\epsilon_{si}/\epsilon_i)t_i$ for high- κ gate dielectrics. However, that is correct only for $t_i \ll \lambda$ when the normal fields dominate. As both ϵ_i and t_i increase by the same factor, i.e., at constant capacitance ϵ_i/t_i , tangential fields become more important for which the high dielectric constant does not help. For arbitrary gate dielectric constant and thickness, it is necessary to apply the generalized two-region scale length model described in Appendix 10. By matching the boundary conditions for both the normal and the tangential fields at the silicon-insulator interface, an eigenvalue equation for the scale length λ is obtained [Eq. (A10.7)]:

$$\frac{1}{\epsilon_i} \tan\left(\frac{\pi t_i}{\lambda}\right) + \frac{1}{\epsilon_{si}} \tan\left(\frac{\pi W_{dm}}{\lambda}\right) = 0. \quad (3.70)$$

This equation has an infinite number of solutions, in descending order of λ . The lowest order eigenvalue or the longest λ dominates because the short-channel potential

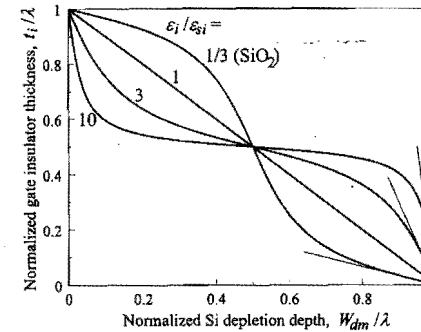


Figure 3.26. Numerical solutions to Eq. (3.70) for different values of ϵ_i/ϵ_{si} . The dotted lines at the lower right corner depict the asymptotic solution behavior, $\lambda \approx W_{dm} + (\epsilon_{si}/\epsilon_i)t_i$, for $t_i \ll W_{dm}$.

component is proportional to $\exp(-\pi L/2\lambda)$ as in the one-region model. Equation (3.70) cannot be solved in closed forms. The numerical solution for the longest λ is shown in normalized units in Fig. 3.26 for several representative values of ϵ_i/ϵ_{si} . **The significance of λ remains that it dictates the minimum channel length, $L_{min} \approx 2\lambda$, as in the one-region case discussed before.**

The following characteristics of the solution to Eq. (3.70) are observed in Fig. 3.26.

- $\lambda \geq W_{dm}$ and $\lambda \geq t_i$, i.e., λ is larger than the larger of W_{dm} , t_i .
- In the special case of $\epsilon_i = \epsilon_{si}$, $\lambda = W_{dm} + t_i$, the physical height of the box in Fig. A10.1.
- In the special case of $W_{dm} = t_i$, $\lambda = 2W_{dm} = 2t_i$, regardless of ϵ_i/ϵ_{si} .
- If $t_i \ll W_{dm}$ (lower right corner of Fig. 3.26), $\lambda \approx W_{dm} + (\epsilon_{si}/\epsilon_i)t_i$. This is the approximate solution obtained in the one-region model.
- If $W_{dm} \ll t_i$ (upper left corner of Fig. 3.26), $\lambda \approx t_i + (\epsilon_i/\epsilon_{si})W_{dm}$.

While Eq. (3.70) and thus Fig. 3.26 are symmetric with respect to t_i and W_{dm} , consideration of $\Delta V_{gs}/\Delta \psi_s$ or the m -factor in Fig. 3.5 requires that $t_i/\epsilon_i < W_{dm}/\epsilon_{si}$. In other words, only the λ solutions in the lower right corner of Fig. 3.26 are acceptable. In that region, high- κ gate dielectric helps because $\lambda \approx W_{dm} + (\epsilon_{si}/\epsilon_i)t_i$ and for the same $W_{dm}/\lambda \lesssim 1$, higher ϵ_i/ϵ_{si} allows a larger t_i/λ . However, because of the extreme nonlinearity of the curves for $\epsilon_i/\epsilon_{si} \gg 1$, the λ solution quickly departs from the above one-region, linear approximation (dotted lines in Fig. 3.26) as t_i/λ increases. There exists a limit of $t_i/\lambda \leq 1/2$, or $\lambda \geq 2t_i$, where t_i is physical thickness of the insulator no matter how high the dielectric constant is. Physically, this is caused by the lateral fields which, unlike the vertical fields, are not affected by the dielectric constant of the material (Section 2.1.4.2). In the devices with thick, very high- κ gate insulators, the short-channel effect is dominated by the lateral fields such that the scale length is determined mainly by the physical thickness of the film.

Notice that for $\epsilon_i/\epsilon_{si} < 1$, e.g., SiO_2 , the curvature of the curve in Fig. 3.26 is opposite to those of $\epsilon_i/\epsilon_{si} > 1$. This means that λ is somewhat lower (better) than the one-region approximation, $W_{dm} + 3t_{ox}$, as t_{ox} increases.

3.2.2 Velocity Saturation and High-Field Transport

As discussed in Section 3.1.2, when the drain voltage increases in a long-channel MOSFET, the drain current first increases, then becomes saturated at a voltage equal to $V_{dsat} = (V_{gs} - V_t)/m$ with the onset of pinch-off at the drain. **In a short-channel device, the saturation of drain current may occur at a much lower voltage due to velocity saturation.** This causes the saturation current I_{dsat} to deviate from the $1/L$ dependence depicted in Eq. (3.28) for long-channel devices. Velocity-field relationships in bulk silicon are plotted in Fig. 2.10. Saturation velocities of electrons and holes in a MOSFET channel are slightly lower than their bulk values. $v_{sat} \approx 7-8 \times 10^6$ cm/s for electrons and $v_{sat} \approx 6-7 \times 10^6$ cm/s for holes have been reported in the literature (Coen and Muller, 1980; Taur *et al.*, 1993a). Figure 3.27 shows the experimentally measured $I_{ds} - V_{ds}$ curves of a 0.25- μm nMOSFET. The dashed curve represents the long-channel-like current given by Eq. (3.28) for $V_{gs} = 2.5$ V. Due to velocity saturation, the drain current saturates at a drain voltage much lower than $(V_{gs} - V_t)/m$, thus severely limiting the saturation current of a short-channel device.

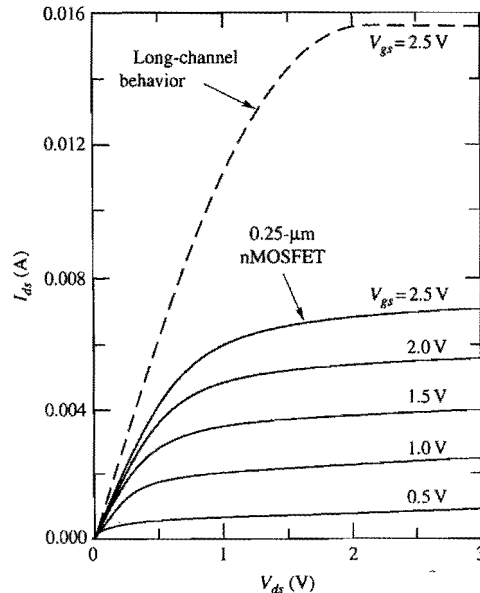


Figure 3.27. Experimental I - V curves of a 0.25- μm nMOSFET (solid lines). The device width is 9.5 μm . The dashed curve shows the long-channel-like drain current expected for this channel length if there were no velocity saturation. (After Taur *et al.*, 1993a.)

3.2.2.1 Velocity-Field Relationship

Experimental measurements show that the velocity-field relationship for electrons and holes takes the empirical form (Caughey and Thomas, 1967)

$$v = \frac{\mu_{eff} \mathcal{E}}{[1 + (\mathcal{E}/\mathcal{E}_c)^n]^{1/n}}, \quad (3.71)$$

where $n=2$ for electrons and $n=1$ for holes. $n (\geq 1)$ is a measure of how rapidly the carriers approach saturation. The parameter $\mathcal{E}_c$ is called the *critical field*. When the field strength is comparable to or greater than $\mathcal{E}_c$, velocity saturation becomes important. At low fields, $v = \mu_{eff} \mathcal{E}$, which is simply Ohm's law. As $\mathcal{E} \rightarrow \infty$, $v = v_{sat} = \mu_{eff} \mathcal{E}_c$. Therefore,

$$\mathcal{E}_c = \frac{v_{sat}}{\mu_{eff}}. \quad (3.72)$$

It was discussed in Section 3.1.5 that the effective mobility μ_{eff} is a function of the vertical (or normal) field $\mathcal{E}_{eff}$. Since v_{sat} is a constant independent of $\mathcal{E}_{eff}$, the critical field $\mathcal{E}_c$ is a function of $\mathcal{E}_{eff}$ as well. More specifically, **for a higher vertical field, the effective mobility is lower, but the critical field for velocity saturation becomes higher** (Sodini *et al.*, 1984). Similarly, holes have a critical field higher than that of electrons, since hole mobilities are lower.

3.2.2.2 An Analytical Solution for $n=1$

It is more important to treat velocity saturation for electrons. However, the mathematics in solving the $n=2$ case is rather tedious (Taylor, 1984). An insight into the velocity saturation phenomenon in a MOSFET can be gained by analyzing the $n=1$ case, which yields a simple and continuous solution. Following similar steps to those in Section 3.1.1, one replaces the low-field drift velocity, $-\mu_{eff} dV/dy$, in Eq. (3.8) with Eq. (3.71) to allow for high-field velocity saturation effects ($n=1$):

$$I_{ds} = -WQ_i(V) \frac{\mu_{eff} dV/dy}{1 + (\mu_{eff}/v_{sat}) dV/dy}. \quad (3.73)$$

Here V is the quasi-Fermi potential at a point y in the channel, and $Q_i(V)$ is the integrated (vertically) inversion charge density at that point. Note that $dV/dy = -\mathcal{E} > 0$.³ Current continuity requires that I_{ds} be a constant, independent of y .

Rearranging Eq. (3.73), one obtains

$$I_{ds} = - \left(\mu_{eff} W Q_i(V) + \frac{\mu_{eff} I_{ds}}{v_{sat}} \right) \frac{dV}{dy}. \quad (3.74)$$

Multiplying by dy on both sides and integrating from $y=0$ to L and from $V=0$ to V_{ds} , one solves for I_{ds} :

³ Strictly speaking, $\mathcal{E} = -d\psi/dy$. Above the threshold, the current is dominated by the drift component; hence $d\psi/dy$ and dV/dy are interchangeable.

$$I_{ds} = \frac{-\mu_{eff}(W/L) \int_0^{V_{ds}} Q_i(V) dV}{1 + (\mu_{eff} V_{ds}/v_{sat} L)} \quad (3.75)$$

The numerator is simply the long-channel current, Eq. (3.10), without velocity saturation. It is clear that if the “average” field along the channel, V_{ds}/L , is much less than the critical field $\mathcal{E}_c = v_{sat}/\mu_{eff}$, the drain current is hardly affected by velocity saturation. When V_{ds}/L becomes comparable to or greater than $\mathcal{E}_c$, however, the drain current is significantly reduced. If one uses the approximate expression (3.29) in the regional charge-sheet model for $Q_i(V)$,

$$Q_i(V) = -C_{ox}(V_{gs} - V_t - mV), \quad (3.76)$$

the integration in Eq. (3.75) can be carried out to yield

$$I_{ds} = \frac{\mu_{eff} C_{ox} (W/L) [(V_{gs} - V_t) V_{ds} - (m/2) V_{ds}^2]}{1 + (\mu_{eff} V_{ds}/v_{sat} L)} \quad (3.77)$$

For a given V_{gs} , I_{ds} increases with V_{ds} until a maximum current is reached. Beyond this point, the drain current is saturated. The saturation voltage, V_{dsat} , can be found by solving $dI_{ds}/dV_{ds} = 0$:

$$V_{dsat} = \frac{2(V_{gs} - V_t)/m}{1 + \sqrt{1 + 2\mu_{eff}(V_{gs} - V_t)/(mv_{sat}L)}} \quad (3.78)$$

This expression is always less than the long-channel saturation voltage, $(V_{gs} - V_t)/m$. Substituting Eq. (3.78) into Eq. (3.77), one finds the saturation current,

$$I_{dsat} = C_{ox} W v_{sat} (V_{gs} - V_t) \frac{\sqrt{1 + 2\mu_{eff}(V_{gs} - V_t)/(mv_{sat}L)} - 1}{\sqrt{1 + 2\mu_{eff}(V_{gs} - V_t)/(mv_{sat}L)} + 1} \quad (3.79)$$

Example curves of I_{dsat} versus $V_{gs} - V_t$ are plotted in Fig. 3.28 for several different channel lengths. In the long-channel case, the solid curve calculated from Eq. (3.79) is not too different from the dashed curve representing the drain current without velocity saturation. In fact, it can be shown that Eq. (3.79) reduces to the long-channel saturation current [Eq. (3.28)],

$$I_{dsat} = \mu_{eff} C_{ox} \frac{W}{L} \frac{(V_{gs} - V_t)^2}{2m}, \quad (3.80)$$

when $V_{gs} - V_t \ll mv_{sat}L/2\mu_{eff}$. As the channel length becomes shorter, the velocity-saturated current (solid curves) is significantly less than that of Eq. (3.80) (dashed curves) over an increasing range of gate voltage. In the limit of $L \rightarrow 0$, Eq. (3.79) becomes the velocity-saturation-limited current,

$$I_{dsat} = C_{ox} W v_{sat} (V_{gs} - V_t), \quad (3.81)$$

as indicated by the straight line labeled $L=0$ in Fig. 3.28. *Note that Eq. (3.81) is independent of channel length L and varies linearly with $V_{gs} - V_t$ instead of*

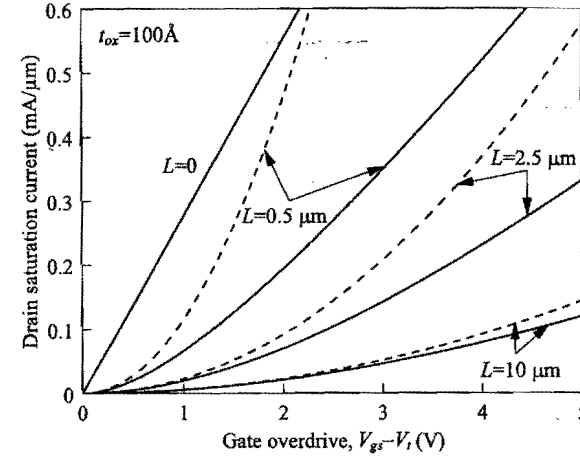


Figure 3.28. Saturation current calculated from Eq. (3.79) versus $V_{gs} - V_t$ for several different channel lengths (solid curves). The dashed curves are the corresponding “long-channel-like” saturation currents calculated from Eq. (3.80), i.e., by letting $v_{sat} \rightarrow \infty$ in Eq. (3.79). The $L=0$ line represents the limiting case imposed by velocity saturation, Eq. (3.81).

quadratically as in the long-channel case. This is consistent with observations of the experimental curves in Fig. 3.27. For very short channel lengths, the saturation voltage, Eq. (3.78), can be approximately by

$$V_{dsat} = \sqrt{2v_{sat}L(V_{gs} - V_t)/m\mu_{eff}}, \quad (3.82)$$

which decreases with channel length.

It is instructive to examine the charge and field behavior at the drain end of the channel when $V_{ds} = V_{dsat}$. From Eq. (3.76),

$$Q_i(y = L) = -C_{ox}(V_{gs} - V_t - mV_{dsat}). \quad (3.83)$$

Substituting V_{dsat} from Eq. (3.78), one finds

$$Q_i(y = L) = -C_{ox}(V_{gs} - V_t) \frac{\sqrt{1 + 2\mu_{eff}(V_{gs} - V_t)/(mv_{sat}L)} - 1}{\sqrt{1 + 2\mu_{eff}(V_{gs} - V_t)/(mv_{sat}L)} + 1} \quad (3.84)$$

Comparison with Eq. (3.79) yields $I_{dsat} = -Wv_{sat}Q_i(y = L)$, i.e., the carrier drift velocity at the drain end of the channel is equal to the saturation velocity. From Eq. (3.73), this means that the lateral field along the channel, dV/dy , approaches infinity at the drain. Just as in the long-channel pinch-off situation discussed in Subsection 3.1.2, such a singularity leads to the breakdown of the gradual-channel approximation which assumed that the lateral field changes slowly in comparison with the vertical field. In other words, *beyond the saturation point, carriers which are traveling at saturation velocity are no longer confined to the surface channel.* Their transport must then be described by a 2-D

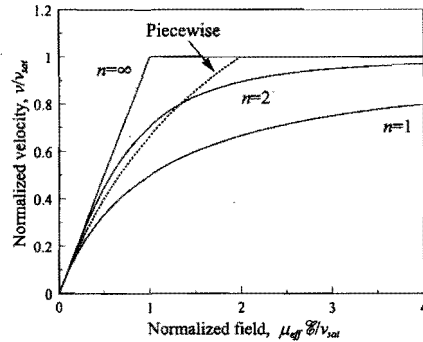


Figure 3.29. Velocity-field relationship of various velocity saturation models plotted in normalized units. The rate of approaching saturation velocity differs in different models.

Poisson equation. A key difference between pinch-off in long-channel devices and velocity saturation in short-channel devices is that in the latter case, the inversion charge density at the drain, Eq. (3.84), does not vanish.

3.2.2.3 $n = \infty$ Velocity Saturation Model

Other than the $n = 1$ velocity saturation model discussed above, analytical solutions also exist in the $n = \infty$ case and a piecewise model depicted in Fig. 3.29. The steepest approach to v_{sat} is obtained by letting $n \rightarrow \infty$ in Eq. (3.71):

$$v = \mu_{eff} \mathcal{E} \quad \text{for} \quad \mathcal{E} < v_{sat} / \mu_{eff}, \quad (3.85)$$

and

$$v = v_{sat} \quad \text{for} \quad \mathcal{E} > v_{sat} / \mu_{eff}. \quad (3.86)$$

For $v < v_{sat}$, the current expression is the same as the long-channel result, Eq. (3.25). In this case, however, before V_{ds} reaches the pinch-off value, $(V_{gs} - V_t)/m$, carrier velocity at the drain end of the channel reaches $v = v_{sat}$ and the current saturates. If this happens at $V_{ds} = V_{dsat}$, then the saturated current is

$$I_{dsat} = \mu_{eff} C_{ox} \frac{W}{L} \left[(V_{gs} - V_t) V_{dsat} - \frac{m}{2} V_{dsat}^2 \right]. \quad (3.87)$$

On the other hand, at the drain end of the channel,

$$I_{dsat} = -WQ_i v_{sat} = WC_{ox} v_{sat} (V_{gs} - V_t - mV_{dsat}), \quad (3.88)$$

where Eq. (3.29) is used for Q_i . Equating Eqs. (3.87) and (3.88) allows V_{dsat} to be solved:

$$V_{dsat} = \frac{V_{gs} - V_t}{m} + \frac{Lv_{sat}}{\mu_{eff}} - \sqrt{\left(\frac{V_{gs} - V_t}{m} \right)^2 + \left(\frac{Lv_{sat}}{\mu_{eff}} \right)^2}. \quad (3.89)$$

Substituting V_{dsat} back into Eq. (3.88) yields the saturation current,

$$I_{dsat} = WC_{ox} v_{sat} \left\{ \sqrt{\left(V_{gs} - V_t \right)^2 + \left(\frac{mLv_{sat}}{\mu_{eff}} \right)^2} - \frac{mLv_{sat}}{\mu_{eff}} \right\}. \quad (3.90)$$

Just like the $n = 1$ case, Eq. (3.90) is also reduced to the long-channel limit, Eq. (3.80), and the fully velocity saturated limit, Eq. (3.81), in the limits of $v_{sat} \rightarrow \infty$ and $L \rightarrow 0$, respectively.

3.2.2.4 A Piecewise-Continuous Velocity Saturation Model

It was mentioned before that while electrons behave like the $n = 2$ model, the analytic solution is too tedious to deal with. A piecewise continuous velocity saturation model was developed instead to approximate the $n = 2$ characteristics (Sodini *et al.*, 1984). At low to moderate fields, the velocity varies with the field like that of an $n = 1$ model except that v_{sat} is replaced with $2v_{sat}$. At high fields, the velocity saturates at v_{sat} once it is reached. In other words,

$$v = \frac{\mu_{eff} \mathcal{E}}{1 + (\mu_{eff} \mathcal{E} / 2v_{sat})} \quad \text{for} \quad \mathcal{E} < 2v_{sat} / \mu_{eff}, \quad (3.91)$$

and

$$v = v_{sat} \quad \text{for} \quad \mathcal{E} > 2v_{sat} / \mu_{eff}. \quad (3.92)$$

The piecewise-continuous model is also shown in Fig. 3.29 (dotted curve). It has been adopted in various BSIM compact models for MOSFETs.

Analytic expressions for the saturation voltage and current can be readily developed for the piecewise model. Before saturation, the drain current is like that of Eq. (3.73) with v_{sat} replaced by $2v_{sat}$. Going through the same derivation as before, one obtains the $I_{ds}(V_{ds})$ results of Eq. (3.77) with v_{sat} replaced by $2v_{sat}$. In particular, at $V_{ds} = V_{dsat}$ when the velocity at the drain reaches v_{sat} , $I_{ds} = I_{dsat}$, i.e.,

$$I_{dsat} = \frac{\mu_{eff} C_{ox} (W/L) [(V_{gs} - V_t) V_{dsat} - (m/2) V_{dsat}^2]}{1 + (\mu_{eff} V_{dsat} / 2v_{sat} L)}. \quad (3.93)$$

I_{dsat} is also related to V_{dsat} by Eq. (3.88) considering velocity saturation at the drain end of the channel. V_{dsat} is then solved by equating Eqs. (3.93) and (3.88):

$$V_{dsat} = \frac{(V_{gs} - V_t)/m}{1 + \mu_{eff}(V_{gs} - V_t)/(2mv_{sat}L)}. \quad (3.94)$$

Substituting V_{dsat} into Eq. (3.88) yields the saturation current (Sodini *et al.*, 1984),

$$I_{dsat} = \frac{\mu_{eff} C_{ox} (W/L) (V_{gs} - V_t)^2 / 2m}{1 + \mu_{eff} (V_{gs} - V_t) / (2m v_{sat} L)} \quad (3.95)$$

Again, Eq. (3.95) is reduced to the long-channel limit, Eq. (3.80), and the fully velocity saturated limit, Eq. (3.81), in the limits of $v_{sat} \rightarrow \infty$ and $L \rightarrow 0$, respectively.

Owing to the piecewise-continuous nature of this model and the $n = \infty$ model, the resulting $I_{ds}(V_{ds})$ curves are also piecewise-continuous. That is, $I_{ds}(V_{ds})$ is continuous, but its derivative is not at the point of saturation. This may cause some problems when they are implemented in compact models for circuit simulation.

3.2.2.5 Velocity Overshoot

All the MOSFET current formulations discussed thus far, including the mobility definition and velocity saturation, are under the realm of the drift-diffusion approximation, which treats carrier transport in some average fashion close to thermal equilibrium with the silicon lattice. **The drift-diffusion model breaks down in ultrashort-channel devices where high field or rapid spatial variation of potential is present.** In such cases, the scattering events are no longer localized, and some fraction of the carriers may acquire much higher than thermal energy over a portion of the device, for example, near the drain. These carriers are not in thermal equilibrium with the silicon lattice and are generally referred to as *hot carriers*. Under these circumstances, it is possible for the carrier velocity to exceed the saturation velocity. This phenomenon is called *velocity overshoot*.

A more rigorous treatment of the carrier transport under spatially nonuniform high-field conditions has been carried out by a Monte Carlo solution of the Boltzmann transport equation for the electron distribution function (Laux and Fischetti, 1988). Figure 3.30 shows the calculated saturation transconductance of nMOSFETs versus channel length, together with experimental results (Sai-Halasz *et al.*, 1988). Local velocity overshoot near the drain starts to occur below 0.2- μm channel length. At channel lengths near 0.05 μm , velocity overshoot takes place over a substantial portion of the device such that the terminal saturation transconductance may exceed the velocity saturation limited value,

$$g_{msat} \equiv \frac{dI_{dsat}}{dV_g} = C_{ox} W v_{sat}, \quad (3.96)$$

from Eq. (3.81).

3.2.2.6 Ballistic MOSFET and Scattering Theory

It should be noted that while the carrier velocity can reach rather high values in the high-field region near the drain, it does not lead to proportionately high currents. This is illustrated in Fig. 3.31 where the band diagram of a MOSFET biased in saturation is shown. At a point near the drain, the average carrier velocity v_d is high, but the inversion charge sheet density $Q_i = C_{ox}(V_{gs} - V_t - mV_{dsat})$ is low. V_{dsat} will assume a value to maintain current continuity such that

$$I_{ds} = WC_{ox} (V_{gs} - V_t - mV_{dsat}) v_d \quad (3.97)$$

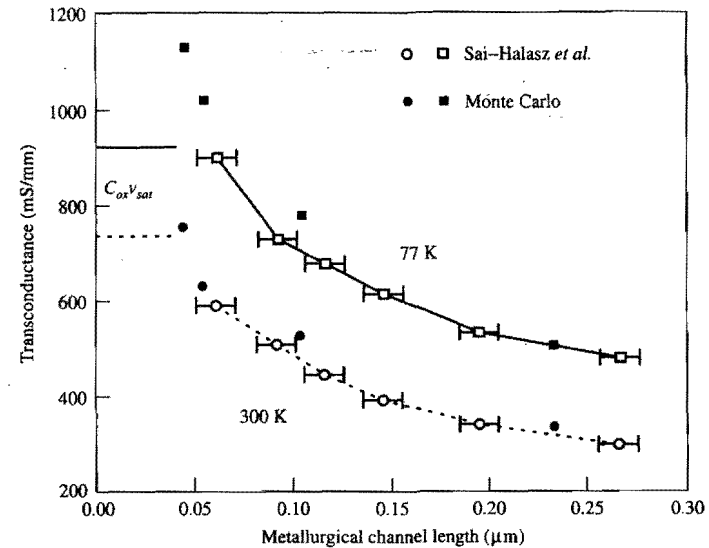


Figure 3.30. Measured (open symbols) and calculated (solid symbols) saturation transconductance versus channel length. The gate oxide is 45 Å thick. Absolute upper bounds for the transconductance in the absence of velocity overshoot at 300 and 77 K are indicated by the lines labeled $C_{ox} v_{sat}$. (After Laux and Fischetti, 1998.)

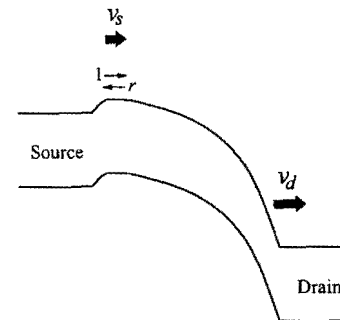


Figure 3.31. Band diagram of a MOSFET biased in saturation; v_s and v_d are the average carrier velocities near the source and the drain, respectively. Carriers near the source can be considered as made up of an incident flux ($1 \rightarrow$) and a reflected flux ($\leftarrow r$) in the scattering theory.

at the drain equals

$$I_{ds} = WC_{ox} (V_{gs} - V_t) v_s \quad (3.98)$$

at the source where $Q_i = C_{ox}(V_{gs} - V_t)$ is high but the carrier velocity v_s is low. In this picture, MOSFET current is more directly related to the average carrier velocity

v_s at the bottleneck region near the source. Velocity overshoot near the drain helps raise MOSFET currents only to the extent that it increases V_{dsat} hence the field near the source.

In the ballistic MOSFET model discussed in Appendix 11 (Natori, 1994), the saturation current is limited by the thermal injection velocity v_T from the source. Equating I_{dsat} of Eq. (A11.11) in the one subband degenerate limit to $WC_{inv}(V_{gs} - V_t)v_T$ yields

$$v_T = \frac{2h}{3\pi m_i} \sqrt{\frac{2C_{inv}(V_{gs} - V_t)}{\pi q}} \quad (3.99)$$

This limit is independent of the field and scattering parameters, and is not enhanced by velocity overshoot near the drain. Note that the ballistic saturation current takes the same form as the velocity-saturation-limited current, Eq. (3.81), with the parameter v_{sat} replaced by the thermal injection velocity v_T . For an electron sheet density of $C_{inv}(V_{gs} - V_t)/q \approx 10^{13} \text{ cm}^{-2}$, $v_T \approx 2 \times 10^7 \text{ cm/s}$, or about twice v_{sat} .

In the scattering theory (Lundstrom, 1997) for an ordinary MOSFET, carriers are injected into the channel from the source (toward the right) and from the drain (toward the left) at their respective injection velocities. They may be back scattered due to random collisions in the channel. Under high drain bias conditions, $V_{ds} \gg kT/q$, carriers originated from the drain have virtually no possibility of making it uphill all the way to the source. Carriers at the point of highest barrier near the source can then be considered as made up of an incident flux (from the source) of amplitude 1 and a reflected flux of amplitude $r < 1$, as shown in Fig. 3.31. Both fluxes are moving at the thermal velocity v_T , but in opposite directions. The average carrier velocity v_s near the source is therefore

$$v_s = \frac{1 - r}{1 + r} v_T. \quad (3.100)$$

The drain current is obtained by substituting v_s in Eq. (3.98). The reflection coefficient r is determined by the field and scattering rates (mobility) in the low-field channel region near the source (Lundstrom, 1997). Once the carriers are a few kT below the highest energy barrier in Fig. 3.31, they are unlikely to be scattered back to the source. In a ballistic MOSFET, there is no scattering, so $r = 0$, $v_s = v_T$, and the current reaches the upper limit.

Note that r is a phenomenological parameter that cannot be predicted from the scattering theory. A Monte Carlo type of solution to the Boltzmann transport equation is needed to calculate r from the detailed physics processes. Recent experimental data suggest that $r \approx 1/3$ and $v_s \approx v_T/2$ in state-of-the-art sub-100 nm MOSFETs (Lochtefeld and Antoniadis, 2001). It has been argued that r is rather insensitive to scaling because of the inevitable loss of mobility (Fig. 3.15) due to increased fields (both lateral and vertical) in the shorter device. This implies that scaling silicon MOSFETs to shorter channel lengths, e.g., 10 nm, may not result in a drain current closer to the ballistic limit.

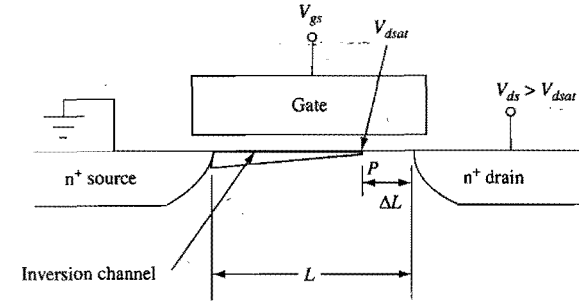


Figure 3.32. Schematic diagram showing channel length modulation when a MOSFET is biased beyond saturation. The surface channel collapses at point P, where carriers reach saturation velocity.

3.2.3 Channel Length Modulation

In this subsection, we discuss the characteristics of short-channel MOSFETs biased beyond saturation. In a long-channel device, the drain current stays constant when the drain voltage exceeds V_{dsat} , as shown in Fig. 3.10. The output conductance, dI_{ds}/dV_{ds} , is zero in the saturation region. In contrast, the drain current of a short-channel MOSFET can still increase slightly beyond the pinch-off or the velocity saturation point with a nonzero output conductance, as is evident from the experimental curves in Fig. 3.27. This arises because of two factors: the short-channel effect and channel length modulation. The short-channel effect was discussed in Section 3.2.1; when the drain voltage increases beyond saturation in a short-channel device, the threshold voltage decreases, and therefore the drain current increases. In this subsection, we describe channel length modulation.

According to the one-dimensional model in the preceding subsection, the electric field along the channel approaches infinity at the saturation point. In practice, the field remains finite. However, its magnitude becomes comparable to the vertical field, so that the gradual-channel approximation breaks down and carriers are no longer confined to the surface channel. As the drain voltage increases beyond the saturation voltage V_{dsat} , the saturation point where the surface channel collapses begins to move slightly toward the source, as shown in Fig. 3.32. The voltage at the saturation point remains constant at V_{dsat} , independent of V_{ds} . The voltage difference $V_{ds} - V_{dsat}$ is dropped across the region between the saturation point and the drain. Carriers injected from the surface channel into this region travel at saturation velocity until collected by the drain junction. The distance between the saturation point and the drain, ΔL , is referred to as the amount of channel length modulation by the drain voltage. Since the one-dimensional model is still valid between the source and the saturation point where the voltage remains at V_{dsat} , the device acts as if its channel length were shortened by ΔL . The drain current is then obtained simply by replacing L with $L - \Delta L$ in Eq. (3.79). In the long-channel limit, this increases the drain current by a factor of $(1 - \Delta L/L)^{-1}$, i.e.,

$$I_{ds} = \frac{I_{dsat}}{1 - (\Delta L/L)}. \quad (3.101)$$

Since ΔL increases with increasing drain voltage, the drain current continues to increase in the saturation region. A 2-D device simulator is needed to numerically evaluate ΔL for a given set of device parameters and bias conditions.

3.2.4 Source-Drain Series Resistance

In the discussion of MOSFET current thus far, it was assumed that the source and drain regions were perfectly conducting. In reality, as the current flows from the channel to the terminal contact, there is a small voltage drop in the source and drain regions due to the finite silicon resistivity and metal contact resistance. In a long-channel device, the source-drain parasitic resistance is negligible compared with the channel resistance. In a short-channel device, however, the source-drain series resistance can be an appreciable fraction of the channel resistance and can therefore cause significant current degradation.

The most severe current degradation by series resistance occurs in the linear region (low V_{ds}) when the gate voltage is high. This is because the MOSFET channel resistance,

$$R_{ch} = \frac{V_{ds}}{I_{ds}} = \frac{L}{\mu_{eff} C_{inv} W (V_{gs} - V_{on})} \quad (3.102)$$

based on Eq. (3.64) and the discussion below it, is the lowest under such bias conditions. It is instructive to estimate the sheet resistivity of a MOSFET channel,

$$\rho_{ch} \equiv R_{ch} \frac{W}{L} = \frac{1}{\mu_{eff} (\epsilon_{ox}/t_{inv})(V_{gs} - V_{on})} \approx \frac{1}{\mu_{eff} \epsilon_{ox} \mathcal{E}_{ox}} \quad (3.103)$$

Since the maximum oxide field $\mathcal{E}_{ox}$ is typically 2–5 MV/cm for most VLSI technologies, the minimum channel sheet resistivity is about 2000 $\Omega/\square$ for nMOSFETs and 7000 $\Omega/\square$ for pMOSFETs.

The MOSFET current in the saturation region is least affected by the resistance degradation of source-drain voltage, since I_{ds} is essentially independent of V_{ds} in saturation. The saturation current is only affected through gate-voltage degradation by the voltage drop between the source contact and the source end of the channel (Fig. 4.23).

The effect of series resistance on linear I_{ds} - V_{gs} curves used in channel-length extraction will be addressed in Section 4.3. Various contributions to the source-drain series resistance and their effect on circuit performance will be discussed in detail in Chapter 5.

3.2.5 MOSFET Degradation and Breakdown at High Fields

Besides causing dielectric breakdown, the high electric fields in a MOSFET can also cause degradation of the device characteristics. *Hot-carrier effects* (HCE) and *negative-bias-temperature instability* (NBTI) are two of the most important degradation phenomena in modern CMOS devices. Here we describe briefly the phenomena and the physical mechanisms involved.

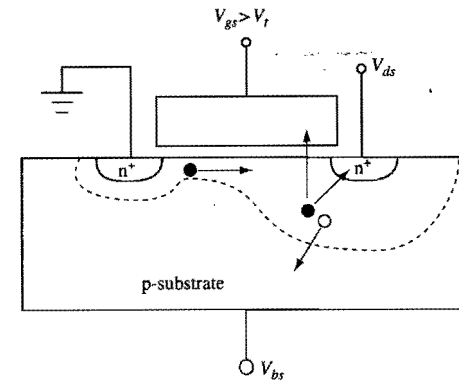


Figure 3.33. Schematic illustrating the physical processes that give rise to channel hot-electron effect in an n-channel MOSFET. The dotted line indicates the boundary of the space-charge region.

3.2.5.1 Hot-Carrier Effects

Consider an n-channel MOSFET with $V_{gs} > V_t$ applied to the gate and V_{ds} applied to the drain. A high-field space-charge region is established in the silicon near the drain, as illustrated in Fig. 3.33. As the electrons drift towards the drain, they gain energy from the electric field in the space-charge region and become hot. The hot electrons can cause impact ionization near the drain, or they can be injected into the gate insulator (see Section 2.5.4). The secondary holes from impact ionization contribute to a substrate current (Abbas, 1974). Substrate currents at drain voltages less than the silicon bandgap voltage E_g/q have been observed, suggesting that some electrons can gain additional energy from electron-electron and/or electron-phonon collisions (Chung *et al.*, 1990).

Figure 3.34 is a typical plot of the channel current and substrate current as a function of the gate voltage, in this case for an n-channel MOSFET with 0.25 μm channel length having a threshold voltage, V_t , of about 0.4 V (Chang *et al.*, 1992). Notice that the substrate current increases with the gate voltage in the subthreshold region, peaking at gate voltages between about V_t and $2V_t$, and then decreases with further increases in the gate voltage. This complex dependence on gate voltage can be understood as follows. The electrons available for initiating impact ionization, to first order, come from the drain current. At $V_{gs} < V_t$, there is no surface inversion channel and the maximum electric field in the silicon near the drain end is relatively independent of the gate voltage. As a result, the substrate current is roughly proportional to the drain current, as can be seen in Fig. 3.34. At $V_{gs} > V_t$, a surface inversion channel is formed. As discussed in Sections 3.1.2.3 and 3.1.2.4, for V_{gs} small compared with V_{ds} , the surface channel is pinched off near the drain end. [Pinch off occurs for $V_{ds} > V_{dsat} = (V_{gs} - V_t)/m$, where m is given by Eq. (3.27).] When the surface inversion channel is pinched off, there is a voltage drop V_{dsat} (drain saturation voltage) along the inversion channel between the source and the pinch-off point, and a voltage drop $V_{ds} - V_{dsat}$ in the space-charge region between the pinch-off point and the drain. That is, the maximum electric field in the silicon

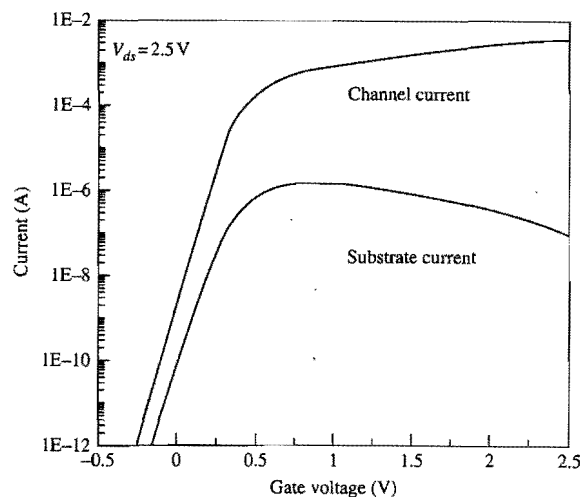


Figure 3.34. Typical plots of the channel current and substrate current of a MOSFET. The example shown here is for an n-channel FET having $0.25\ \mu\text{m}$ channel length and $10\ \mu\text{m}$ channel width. (After Chang *et al.*, 1992.)

space-charge region is determined by $V_{ds} - V_{dsat}$. For a given V_{ds} , as V_{gs} increases, so does V_{dsat} , which means the voltage drop $V_{ds} - V_{dsat}$ decreases hence the maximum electric field in the silicon decreases. As shown in Fig. 2.59, the rate of impact ionization decreases rapidly with decrease in electric field. The net result is that the substrate current decreases with increasing gate voltage for V_{gs} larger than about $2V_t$, as seen in Fig. 3.34. The reader interested in detailed models for the substrate current in a MOSFET is referred to the literature (e.g., Hu *et al.*, 1985, and Kolhatkar and Dutta, 2000).

The hot electrons injected into the gate insulator near the drain region contribute to a gate current. The gate current can cause bulk and interface traps to be generated (see Section 2.5.3.3), and some of the injected electrons can become trapped in the gate insulator near the drain. The trapped electrons and the interface states cause the surface potential near the drain to shift. Since the source of the hot electrons is the channel current, this device degradation is referred to as *channel hot electron (CHE) effect*. The turn on characteristics of a MOSFET are determined primarily by the surface potential near the source end of the channel (see Section 3.1.2). With the damage localized near the drain junction, a MOSFET that has suffered significant CHE damage shows a larger damage-induced threshold voltage shift when it is operated in the source-drain-reversed mode than in the normal mode (Abbas and Dockerty, 1975; Ning *et al.*, 1977b). In the case of a p-channel MOSFET, the same device degradation is referred to as *channel hot hole effect*.

In certain circuit configurations, the terminal voltages of a MOSFET are such that V_{ds} is small or zero, while both the source and the drain are reverse biased with respect to the substrate, i.e., $V_{hs} < 0$. (An example of this bias configuration is when a MOSFET is used as a pass-gate.) In this case, the channel electrons do not gain much energy traveling

from source to drain. However, the minority electrons from the p-substrate can gain energy as they drift towards the silicon-oxide interface. These hot electrons can be injected into the gate insulator as depicted in Fig. 2.68. The injected electrons can generate bulk and interface traps or become trapped in the oxide layer. Since the hot electrons come from the substrate, the associated device degradation is referred to as *substrate hot electron (SHE) effect*. By symmetry, SHE damage to the oxide is spread more or less uniformly over the entire device channel region. Also, since the minority electron density in the substrate increases with temperature, SHE effect increases with temperature (Ning *et al.*, 1976).

In the case of a p-channel MOSFET, instead of hot electrons, we have hot holes causing device degradation. Hot-carrier degradation is one of the major effects limiting the voltage that can be applied to CMOS devices. In general, the approach to minimize hot-carrier damage is to (a) reduce the peak electric field in the silicon to reduce the energy of hot carriers, and (b) minimize the trap density in the gate insulator. Designers often employ the so-called *lightly doped drain (LDD)* design to suppress CHE effect (Ogura *et al.*, 1982). In an LDD design, the drain region adjacent to the channel has a lower doping concentration than the drain region away from the channel. This laterally graded drain doping profile reduces the peak electric field near the drain. As for the substrate hot electron or substrate hot hole effects, they can be suppressed quite effectively by using a relatively lightly doped substrate (Ning *et al.*, 1979). However, both the trap density of the gate insulator and the susceptibility of the gate insulator to hot-carrier damage depend on the growth process of the gate insulator, as well as on the subsequent processes used to complete the integrated-circuit chip fabrication. As a result, hot-carrier effects cannot be predicted sufficiently accurately in advance. Instead, for each CMOS technology generation, the effects are characterized, modeled and then included in the design of circuits. The reader is referred to the literature on the subject for more details (see, e.g., Takeda *et al.*, 1995).

3.2.5.2 Negative-Bias-Temperature Instability

It was reported by Deal *et al.* (1967) that a negative voltage applied to the gate of an MOS capacitor at elevated temperatures can cause both a build up of positive charge in the oxide and an increase in the density of surface states (see Fig. 2.52). To turn on a p-channel MOSFET, the gate electrode is biased negatively with respect to the n-type body. Also, in many applications, the device temperature can be rather high, often close to $100\ ^\circ\text{C}$. Thus, NBTI can occur naturally in the operation of a p-channel MOSFET, causing the gate voltage needed to turn on the transistor to increase (becoming more negative with respect to the n-type body) with time. Many papers have been published on the various aspects of NBTI. It is found that device degradation due to NBTI is a function of the gate insulator process, and the degradation worsens with both the temperature and the oxide field (Jeppson and Svensson, 1977; Blat *et al.*, 1991). For short time periods, the degradation has approximately a $t^{1/4}$ time dependence (Jeppson and Svensson, 1977). It then saturates to a value depending on the oxide field and the temperature (Zafar *et al.*, 2004).

If a p-channel MOSFET is stressed with zero or relatively small voltages between the source and drain, there is only NBTI effect and the channel hot hole effect is negligible. In

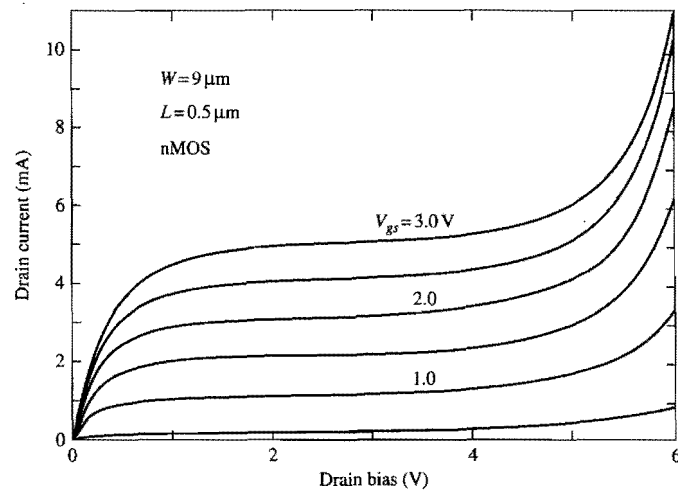


Figure 3.35. Example $I_{ds} - V_{ds}$ curves of a short-channel nMOSFET showing breakdown at high drain voltages. (After Sun *et al.*, 1987.)

this stress mode, the device degradation or NBTI by itself is relatively insensitive to channel length. However, a pMOSFET in the off state in a CMOS circuit typically has a relatively large voltage between its source and drain. Therefore, the device degradation experienced by a short-channel pMOSFET is often caused by a combination of channel hot hole effect and NBTI (La Rosa *et al.*, 1997). NBTI must be characterized for each technology so its effect can be included in the design of circuits, particularly for CMOS circuits that depend on good matching of the threshold voltage of p-channel MOSFETs (Rauch III, 2002). An excellent review of the current understanding of the physical mechanisms of NBTI in modern CMOS devices has been given by Schroder and Babcock (Schroder and Babcock, 2003).

3.2.5.3 MOSFET Breakdown

Breakdown occurs in a short-channel MOSFET when the drain voltage exceeds a certain value, as shown in Fig. 3.35. At high drain voltages, the peak electric field in the saturation region can attain large values. When the field exceeds mid- 10^5 V/cm, impact ionization (Section 2.5.1) takes place at the drain, leading to an abrupt increase of drain current. The breakdown voltage of nMOSFETs is usually lower than that of pMOSFETs because electrons have a higher rate of impact ionization (Fig. 2.59) and because n^+ source and drain junctions are more abrupt than p^+ junctions. There is also a weak dependence of the breakdown voltage on channel length; shorter devices have a lower breakdown voltage.

The breakdown process in an nMOSFET is shown schematically in Fig. 3.36. Electrons gain energy from the field as they move down the channel. Before they lose energy through collisions, they possess high kinetic energy and are capable of generating secondary electrons and holes by impact ionization. The generated electrons are attracted

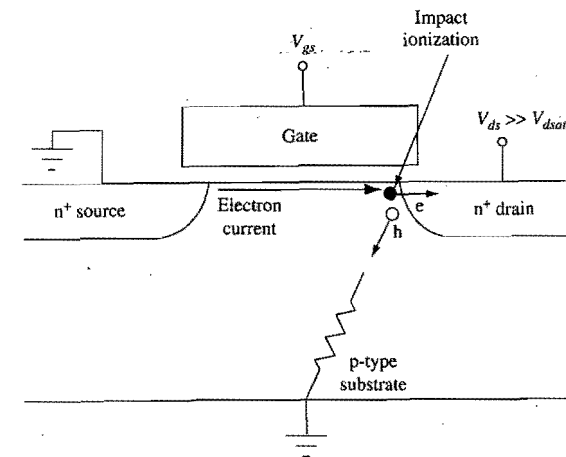


Figure 3.36. Schematic diagram showing impact ionization at the drain.

to the drain, adding to the drain current, while the holes are collected by the substrate contact, resulting in a substrate current. The substrate current in turn can produce a voltage (IR) drop from the spreading resistance in the bulk, which tends to forward-bias the source junction. This lowers the threshold voltage of the MOSFET and triggers a positive feedback effect, which further enhances the channel current. Substrate current (Fig. 3.34) is usually a good indicator of hot carriers generated by low-level impact ionization before runaway breakdown occurs.

Breakdown often results in permanent damage to the MOSFET as large amounts of hot carriers are injected into the oxide in the gate-to-drain overlap region. MOSFET breakdown is particularly a problem for VLSI technology during the elevated-voltage burn-in process. It can be relieved to some extent by using a lightly doped drain (LDD) structure (Ogura *et al.*, 1982), which introduces additional series resistance and reduces the peak field in a MOSFET. However, drain current and therefore device performance are traded off as a result. Ultimately, the devices should operate at a power-supply voltage far enough below the breakdown condition. This is one of the key CMOS design considerations in Chapter 4.

Exercises

- 3.1 Consider an n-channel MOSFET with 20 nm thick gate oxide and uniform p-type substrate doping of 10^{17} cm^{-3} . The gate work function is that of n^+ Si.
 - (a) What is the threshold voltage? Sketch the band diagram at threshold condition, $\psi_s = 2\psi_B$.
 - (b) What is the threshold voltage if a reverse bias of 1 V is applied to the substrate? Sketch the band diagram at threshold.
 - (c) What is the scale length of this device and how short can the channel length be reduced to before severe short-channel effect takes place?

- 3.2 Fill in the steps that lead to Eq. (3.34), the fraction of drift current component in the limit of $V \rightarrow 0$.
- 3.3 The effective field $\mathcal{E}_{eff}$ plays an important role in MOSFET channel mobility. Show that the definition

$$\mathcal{E}_{eff} \equiv \frac{\int_0^{x_i} n(x) \mathcal{E}(x) dx}{\int_0^{x_i} n(x) dx}$$

leads to Eq (3.51), i.e., $\mathcal{E}_{eff} = (|Q_d| + |Q_i|/2)/\epsilon_{si}$. Note that

$$|Q_i| = q \int_0^{x_i} n(x) dx$$

and

$$\mathcal{E}(x) = \frac{1}{\epsilon_{si}} \left(|Q_d| + q \int_x^{x_i} n(x') dx' \right)$$

from Gauss's law. The inversion-layer depth x_i is assumed to be much smaller than the bulk depletion width.

- 3.4 An alternative threshold definition is based on the rate of change of inversion charge density with gate voltage. Equation (3.61) from Fig. 2.37(b) states that $d|Q_i|/dV_{gs}$ is given by the serial combination of C_{ox} and $C_i \equiv d|Q_i|/d\psi_s$. Below threshold, $C_i \ll C_{ox}$, so that $d|Q_i|/dV_{gs} \approx C_i$ and Q_i increases exponentially with V_{gs} . Above threshold, $C_i \gg C_{ox}$, so that $d|Q_i|/dV_{gs} \approx C_{ox}$ and Q_i increases linearly with V_{gs} . The change of behavior occurs at an *inversion charge threshold voltage*, V_{t}^{inv} , where $C_i = C_{ox}$. Show that at $V_{gs} = V_{t}^{inv}$ one has $d|Q_i|/dV_{gs} = C_{ox}/2$ and $Q_i \approx (2kT/q)C_{ox}$. Note that such an inversion charge threshold is independent of depletion charge and is slightly higher than the conventional $2\psi_B$ threshold.
- 3.5 From Eq. (3.63) (neglecting the second term from inversion charge capacitance), show that the fractional loss of inversion charge due to the polysilicon depletion effect is $\Delta Q_i/Q_i \approx C_{ox}/2C_p$ where C_p is the small-signal polysilicon-depletion capacitance defined in Eq. (2.212). Explain why there is a factor-of-two difference between the loss of charge and the loss of capacitance.
- 3.6 For an nMOSFET with $t_{ox} = 10$ nm, $\mu_{eff} = 500$ cm²/V-s, $v_{sat} = 10^7$ cm/s, $W = 10$ μ m, and $L = 1$ μ m, assume $m = 1$.
- Use the $n = 1$ velocity saturation model to generate I_{ds} versus V_{ds} (0–5 V) curves for $V_{gs} - V_t = 1, 2, 3, 4$, and 5 V. (Note: $I_{ds} = I_{dsat}$ beyond V_{dsat} .)
 - Now let L vary from 0.5 μ m to 5 μ m. Calculate and plot the saturation current for $V_{gs} - V_t = 3$ V vs. L . Compare it with the long-channel saturation current (without velocity saturation) for the same $V_{gs} - V_t$ and range of L .
- 3.7 The small-signal transconductance in the saturation region is defined as $g_{msat} \equiv dI_{dsat}/dV_{gs}$. Derive an expression for g_{msat} using Eq. (3.79) based on the $n = 1$ velocity saturation model. Show that g_{msat} approaches the saturation-velocity-limited value, Eq. (3.96), when $L \rightarrow 0$. What becomes of the expression for g_{msat} in the long-channel limit when $v_{sat} \rightarrow \infty$?

- 3.8 From Eq. (3.79) based on the $n = 1$ velocity saturation model, what is the carrier velocity at the source end of the channel? What are the limiting values when $L \rightarrow 0$ and when $v_{sat} \rightarrow \infty$?
- 3.9 Following a similar approach as in the text for the $n = 1$ velocity saturation model, derive an integral equation for the $n = 2$ velocity saturation model from which I_{ds} can be solved. It is very tedious to carry out the integration analytically (Taylor, 1984). Interested readers may attempt performing it numerically on a computer.
- 3.10 Assuming the $n = 1$ velocity saturation model, show that the total integrated inversion charge under the gate is

$$Q_i(\text{total}) = -WLC_{ox}(V_{gs} - V_t) \frac{\sqrt{1 + 2\mu_{eff}(V_{gs} - V_t)/(mv_{sat}L)} + \frac{1}{3}}{\sqrt{1 + 2\mu_{eff}(V_{gs} - V_t)/(mv_{sat}L)} + 1}$$

in the saturation region. Evaluate the intrinsic gate-to-channel capacitance, and show that it approaches Eq. (3.60) in the long-channel limit.

- 3.11 The generalized MOSFET scale length is given by Eq. (3.70). For $\epsilon_{si} = 11.7\epsilon_0$, $\epsilon_i = 7.8\epsilon_0$, $t_i = 5.0$ nm, and $W_{dm} = 10.0$ nm, find the three longest eigenvalues $\lambda_1, \lambda_2, \lambda_3$. Take $L \approx 2\lambda_1$; what's the ratio between $\exp(-\pi L/2\lambda_1)$ and $\exp(-\pi L/2\lambda_2)$?